

Logistic Regression

Topics covered

After completing this topic, you should be able to:

- Explain why linear regression is unsuitable for classification.
- Describe the sigmoid function and logistic regression model.
- Calculate class probabilities manually.
- Apply a decision threshold to predict classes.
- Explain the log-loss function.
- Interpret logistic regression coefficients and odds ratios.
- Implement logistic regression using Python.
- Evaluate the model using classification metrics.

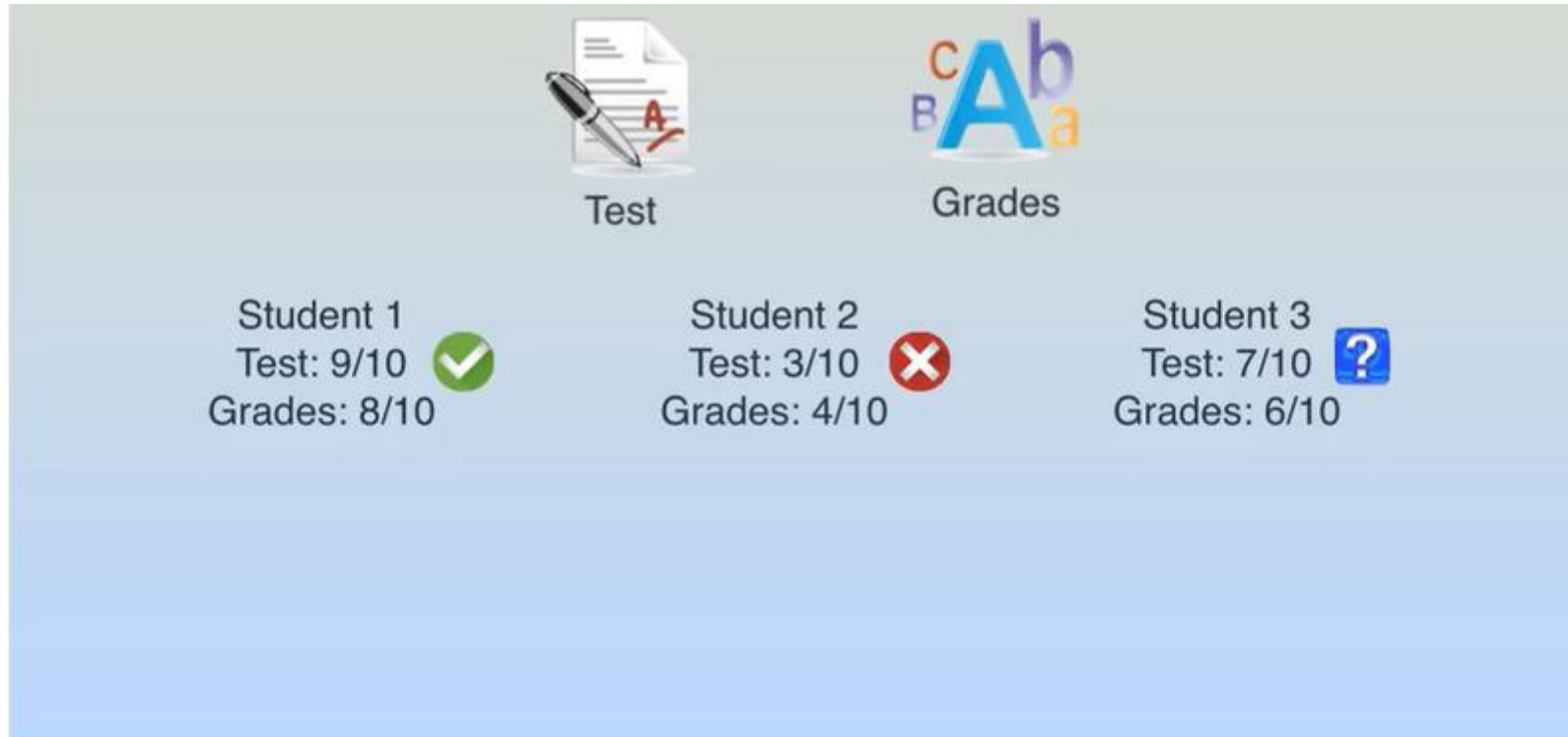
Introduction

- Logistic regression is a **supervised machine-learning** algorithm mainly used for **classification** problems.

Examples:

- Email: spam or not spam
- Loan application: approved or rejected
- Patient: disease or no disease
- Customer: purchases or does not purchase
- Student: pass or fail

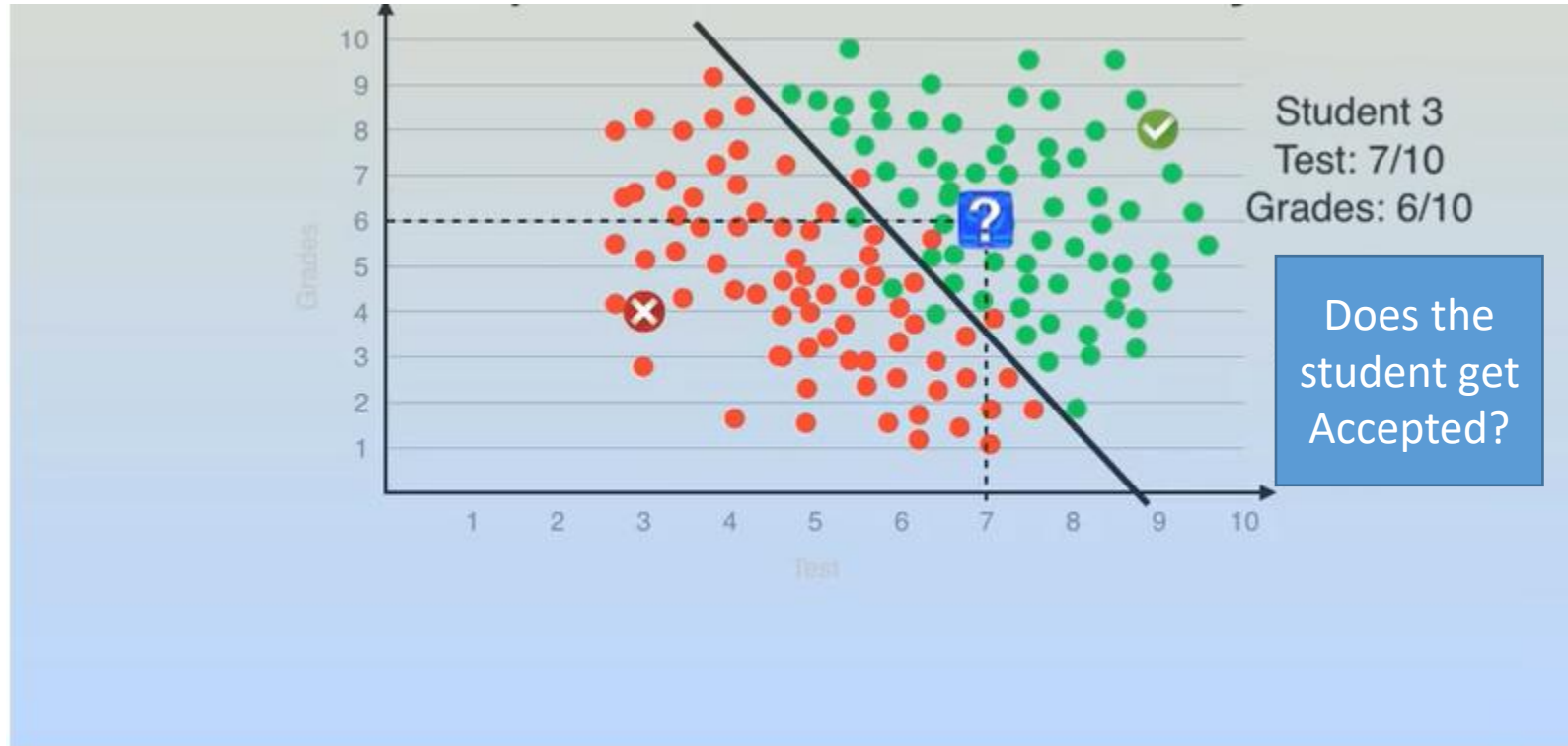
Students' Acceptance at a University



Students' Acceptance at a University



Logistic Regression



Logistic regression

- Logistic regression is one of the most popular machine learning algorithms for [classification](#).
- It is a simple algorithm that performs very well on a wide range of problems.

Type	Description	Example
Binary logistic regression	Two output classes	Pass/Fail
Multinomial logistic regression	More than two unordered classes	Bus/Car/Bike
Ordinal logistic regression	Ordered output classes	Low/Medium/High

Why Not Use Linear Regression?

- Suppose we use linear regression for a binary classification problem:

$$\hat{y} = \beta_0 + \beta_1 x = b_0 + b_1 x$$

- Linear regression can generate predictions such as:

$$\hat{y} = -0.4 \quad \text{or} \quad \hat{y} = 1.3$$

- However, probabilities must lie between 0 and 1.
- Logistic regression solves this problem by transforming the linear output using the **sigmoid function**.

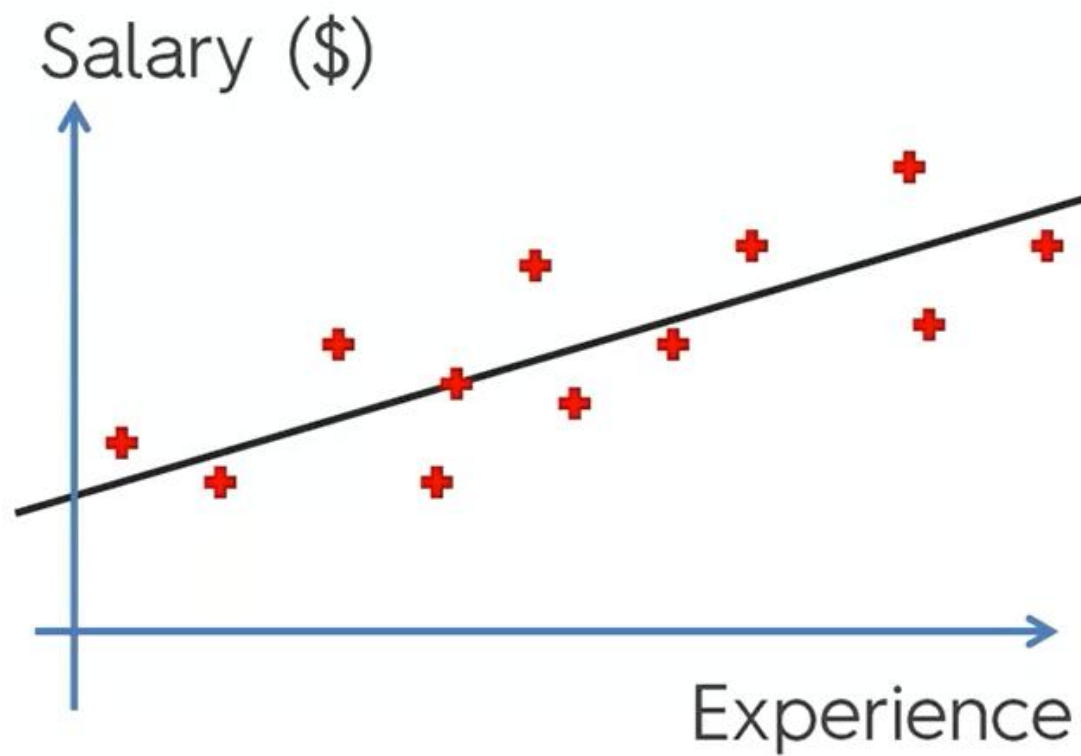
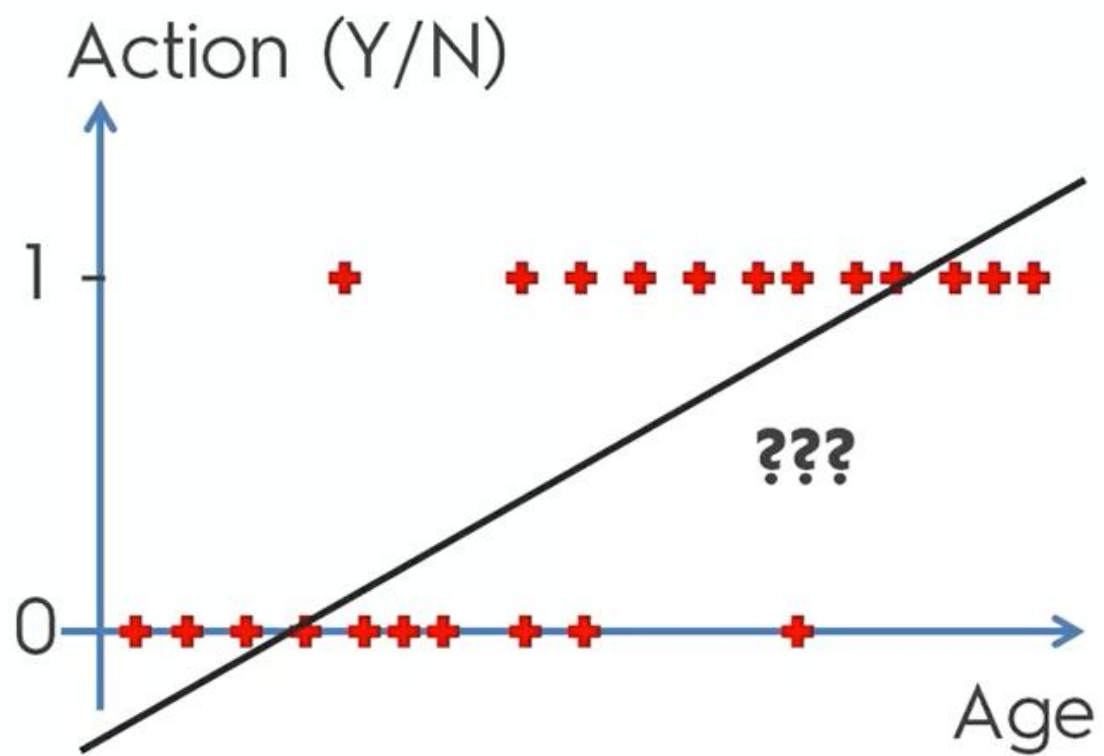
Linear Regression:

- **Simple:**

$$y = b_0 + b_1 * x$$

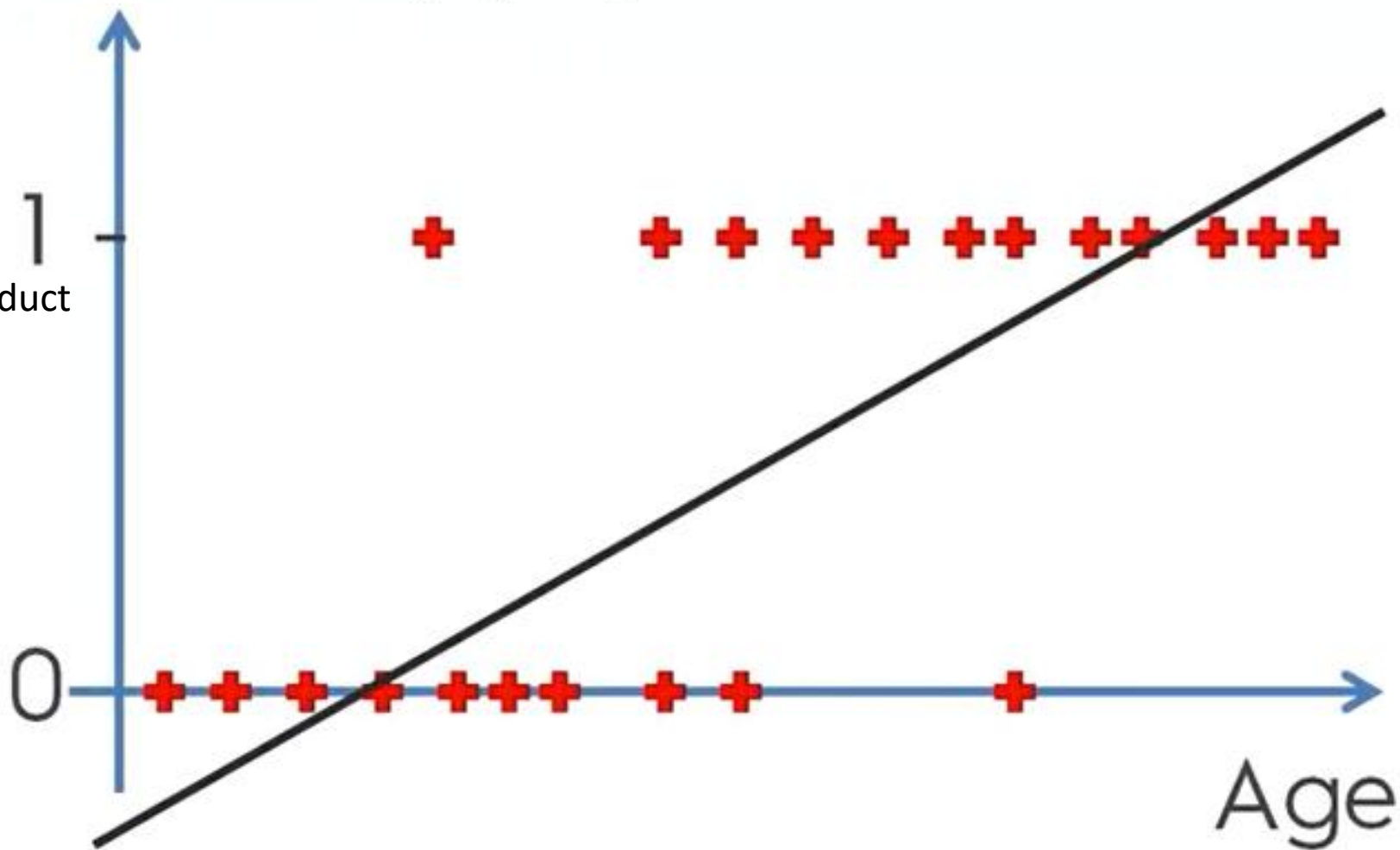
- **Multiple:**

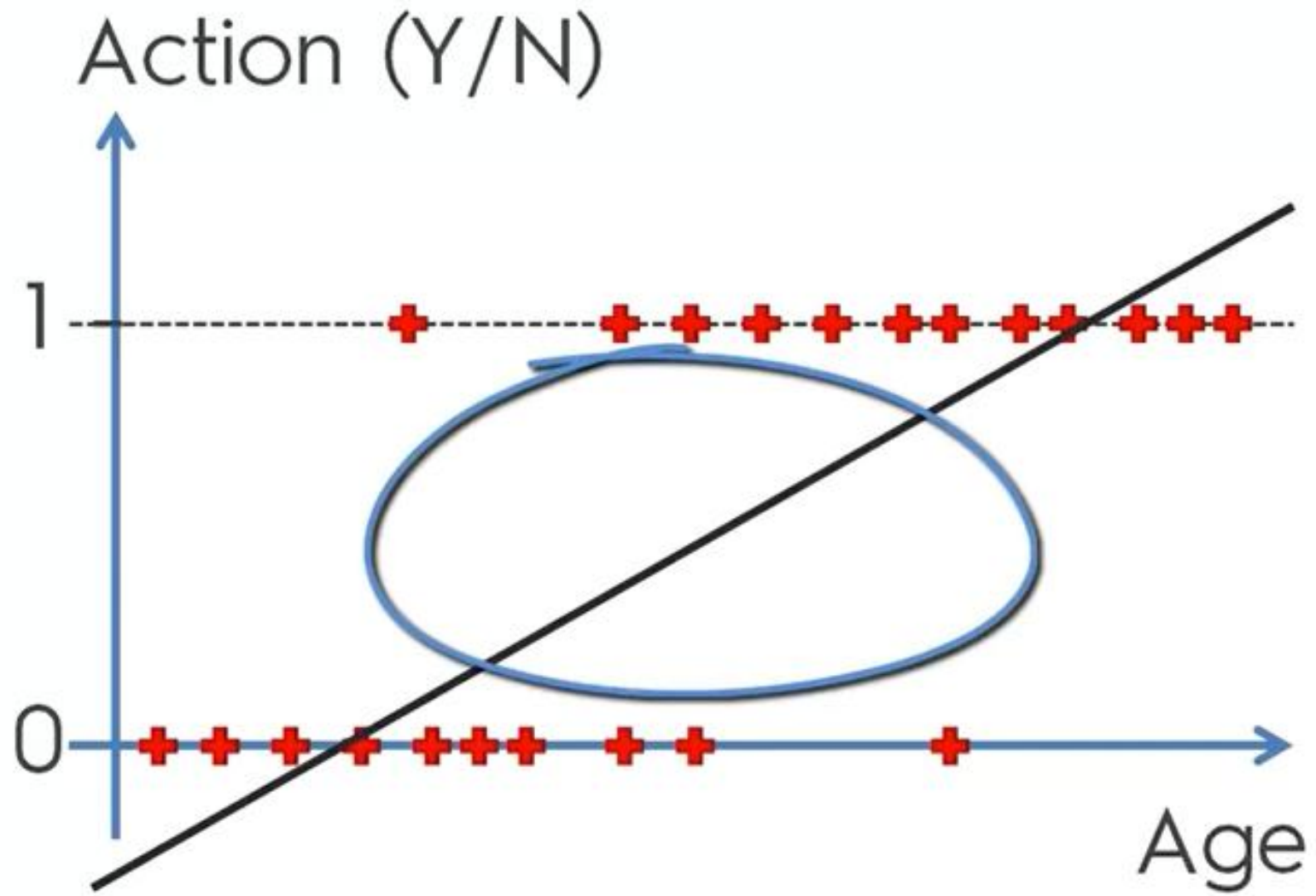
$$y = b_0 + b_1 * x_1 + ... + b_n * x_n$$

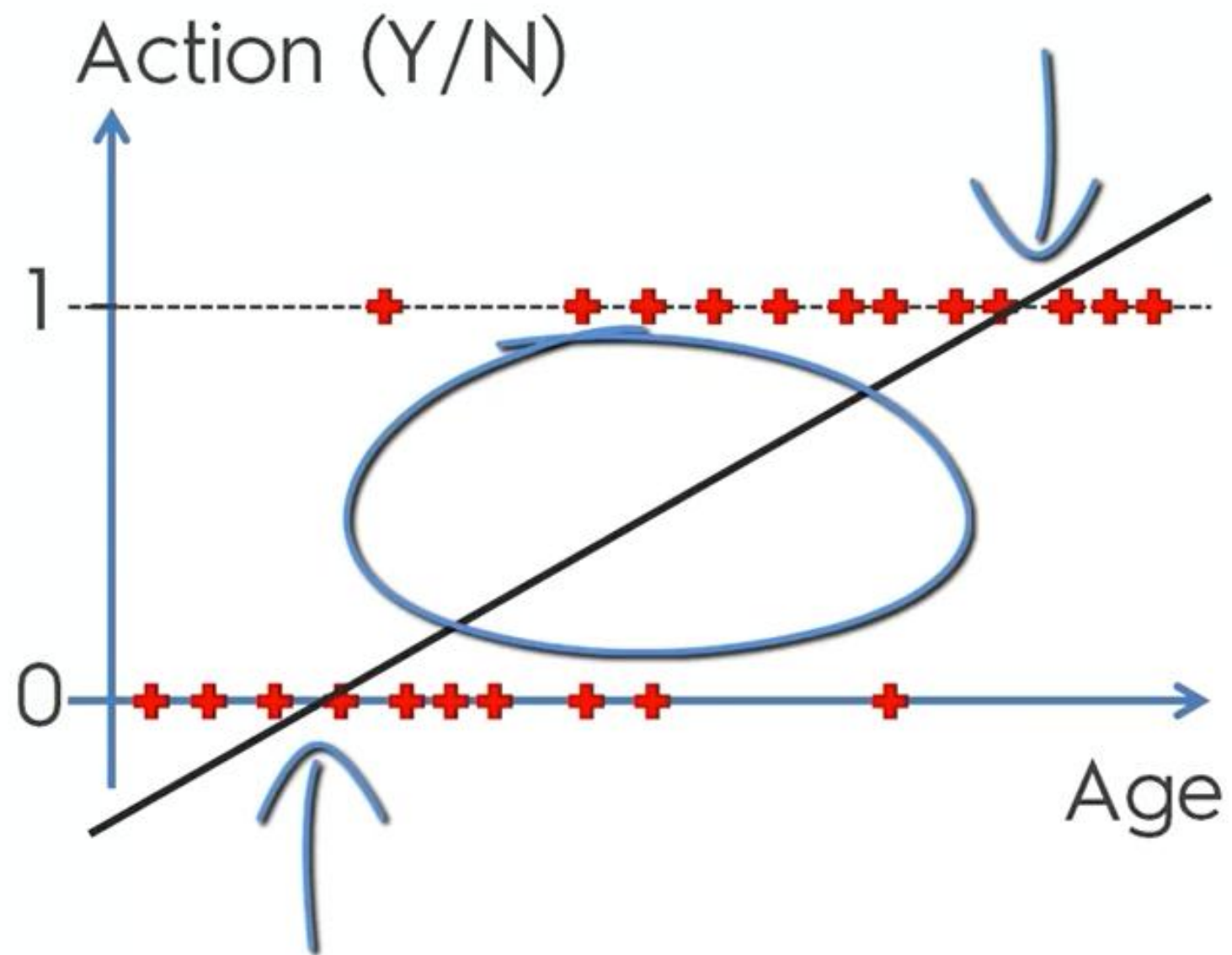


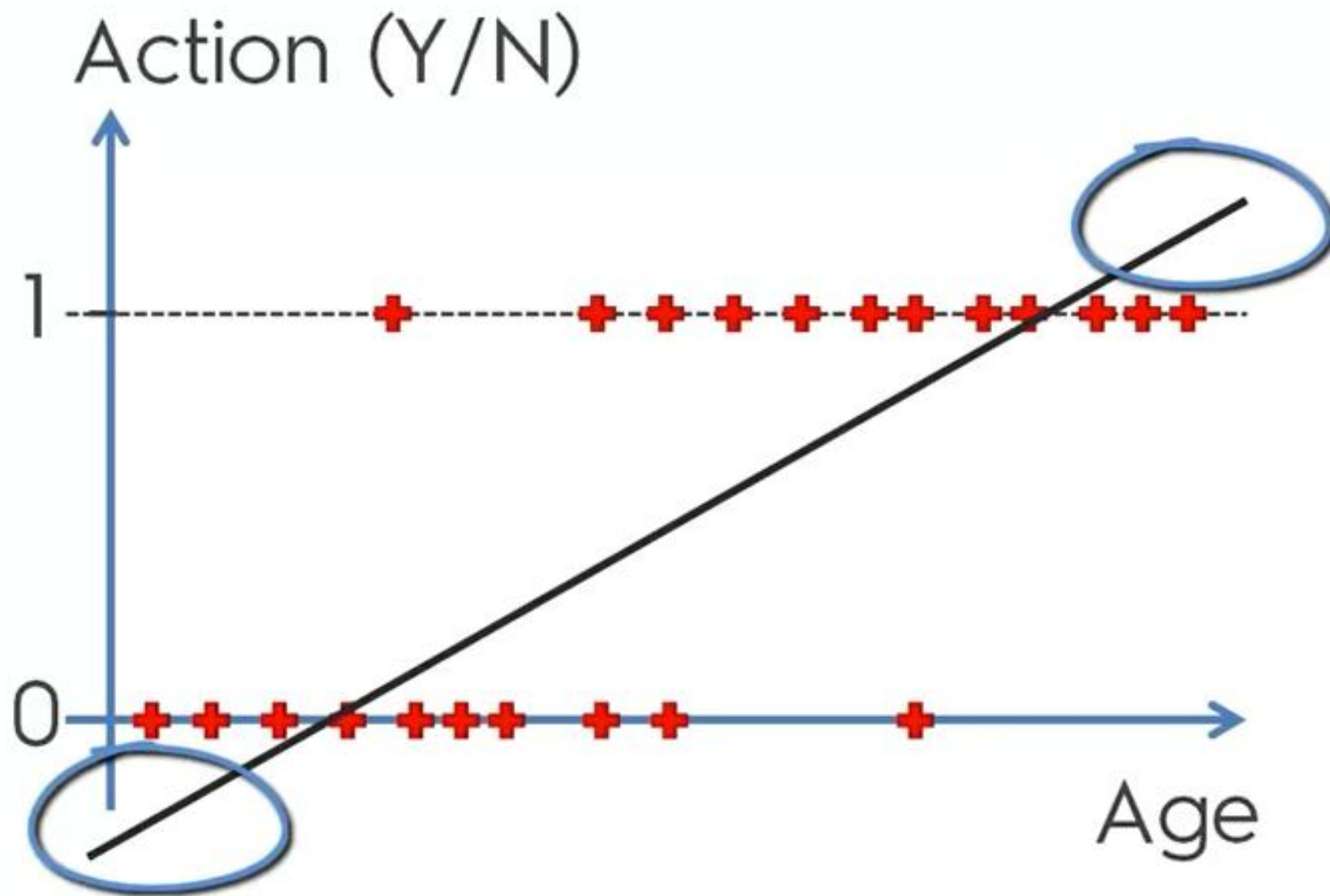
Action (Y/N)

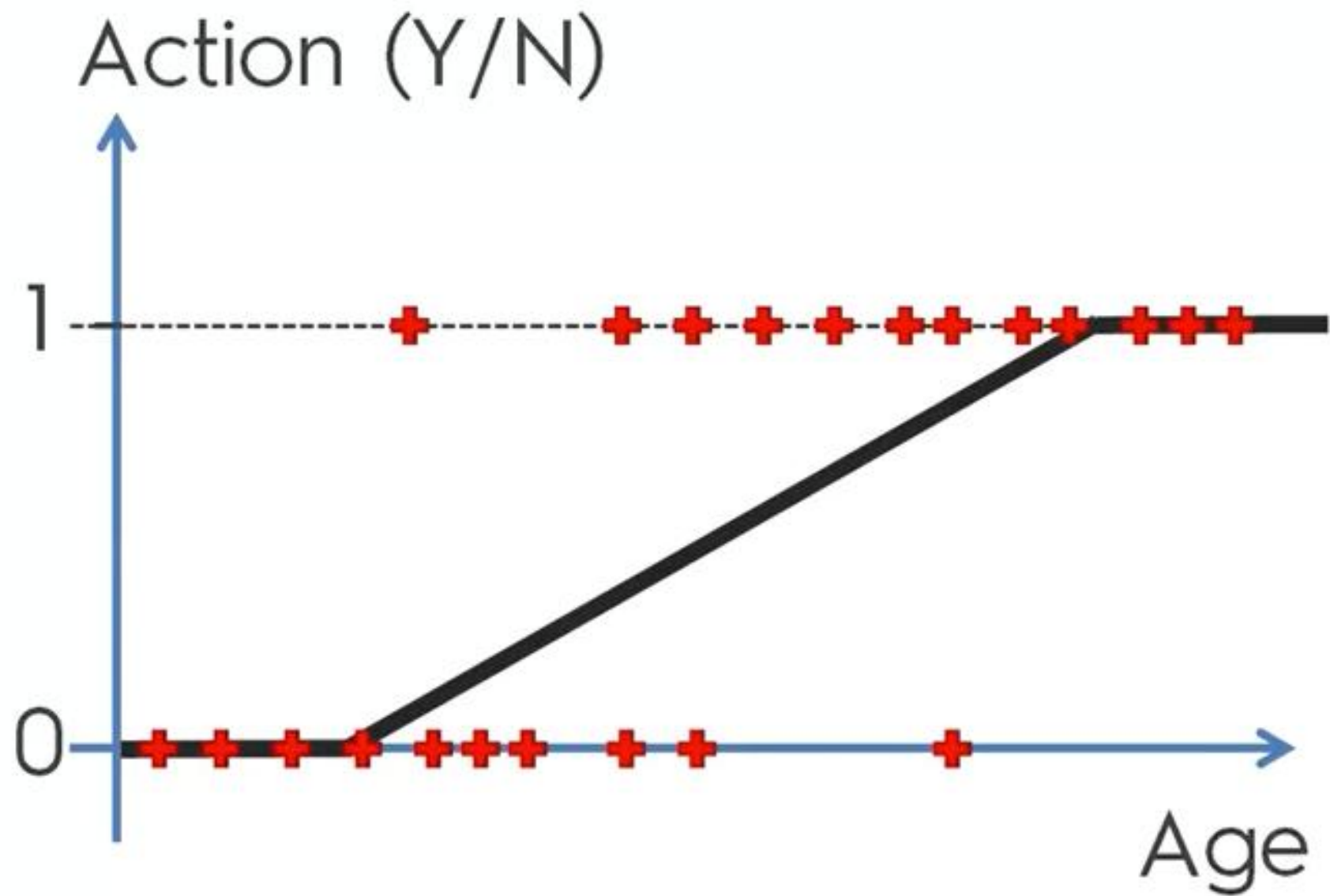
If we consider the
probability
of purchase of product



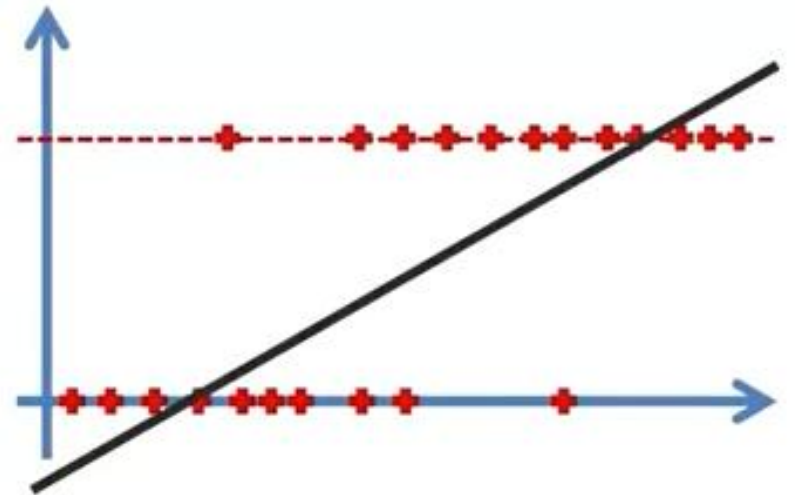








$$y = b_0 + b_1 * x$$

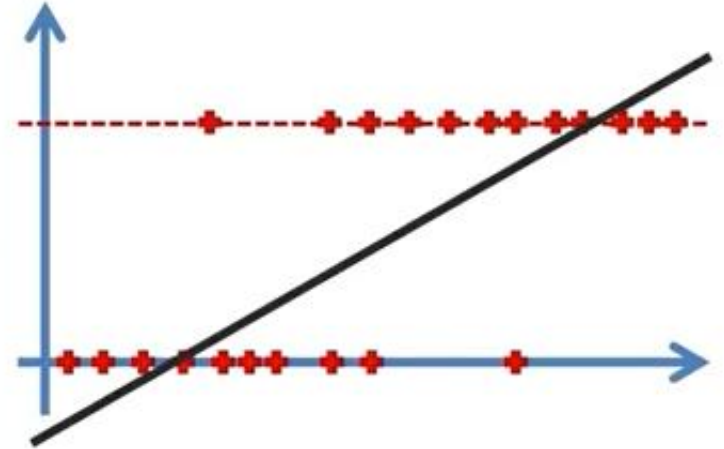


$$y = b_0 + b_1 * x$$



Sigmoid Function

$$p = \frac{1}{1 + e^{-y}}$$

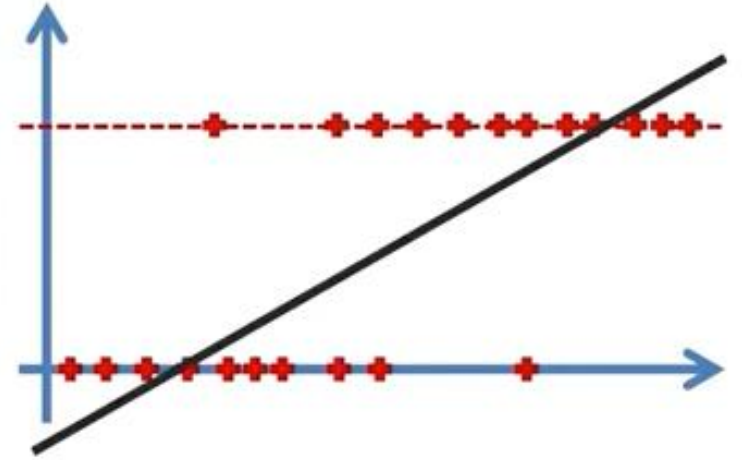


$$y = b_0 + b_1 * x$$

Sigmoid Function

$$p = \frac{1}{1 + e^{-y}}$$

$$\ln \left(\frac{p}{1 - p} \right) = b_0 + b_1 * x$$

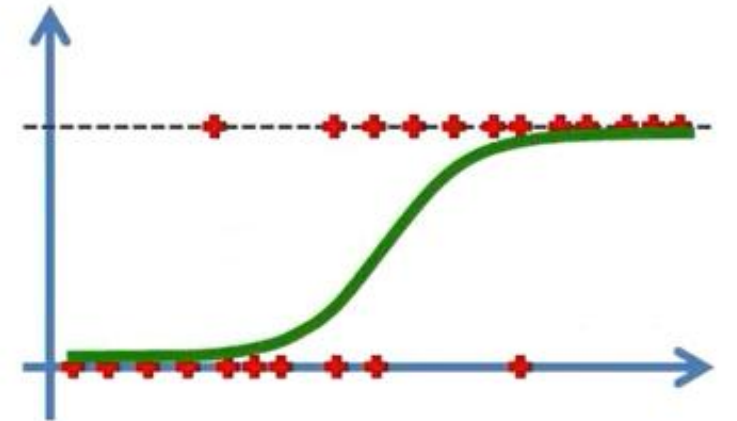
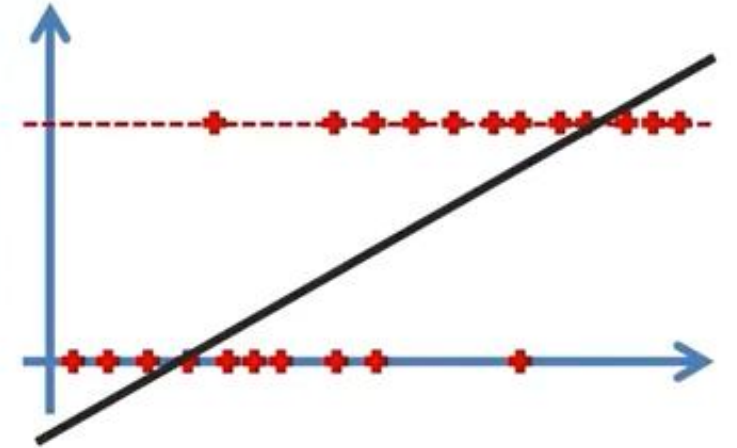


$$y = b_0 + b_1 * x$$

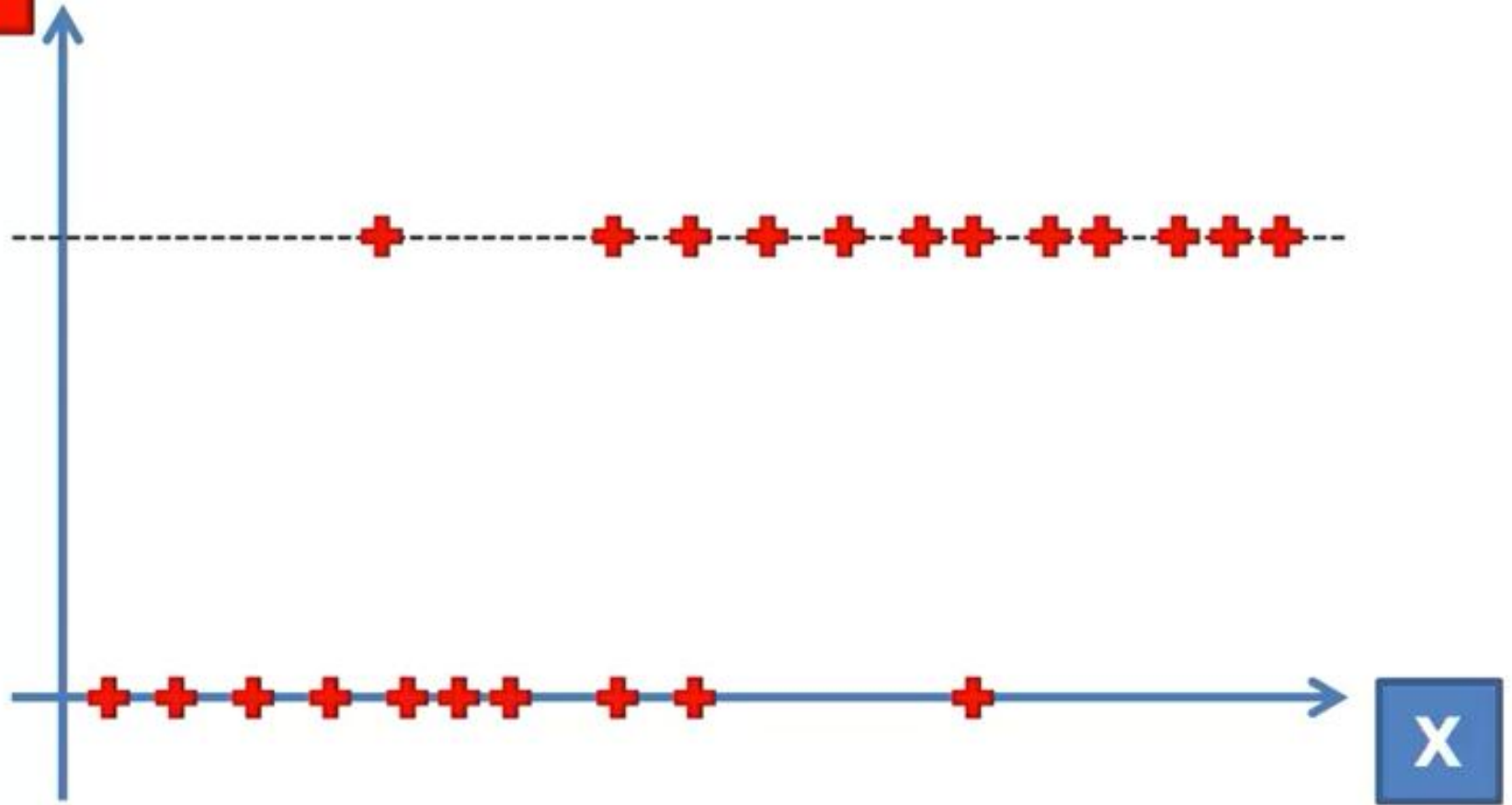
Sigmoid Function

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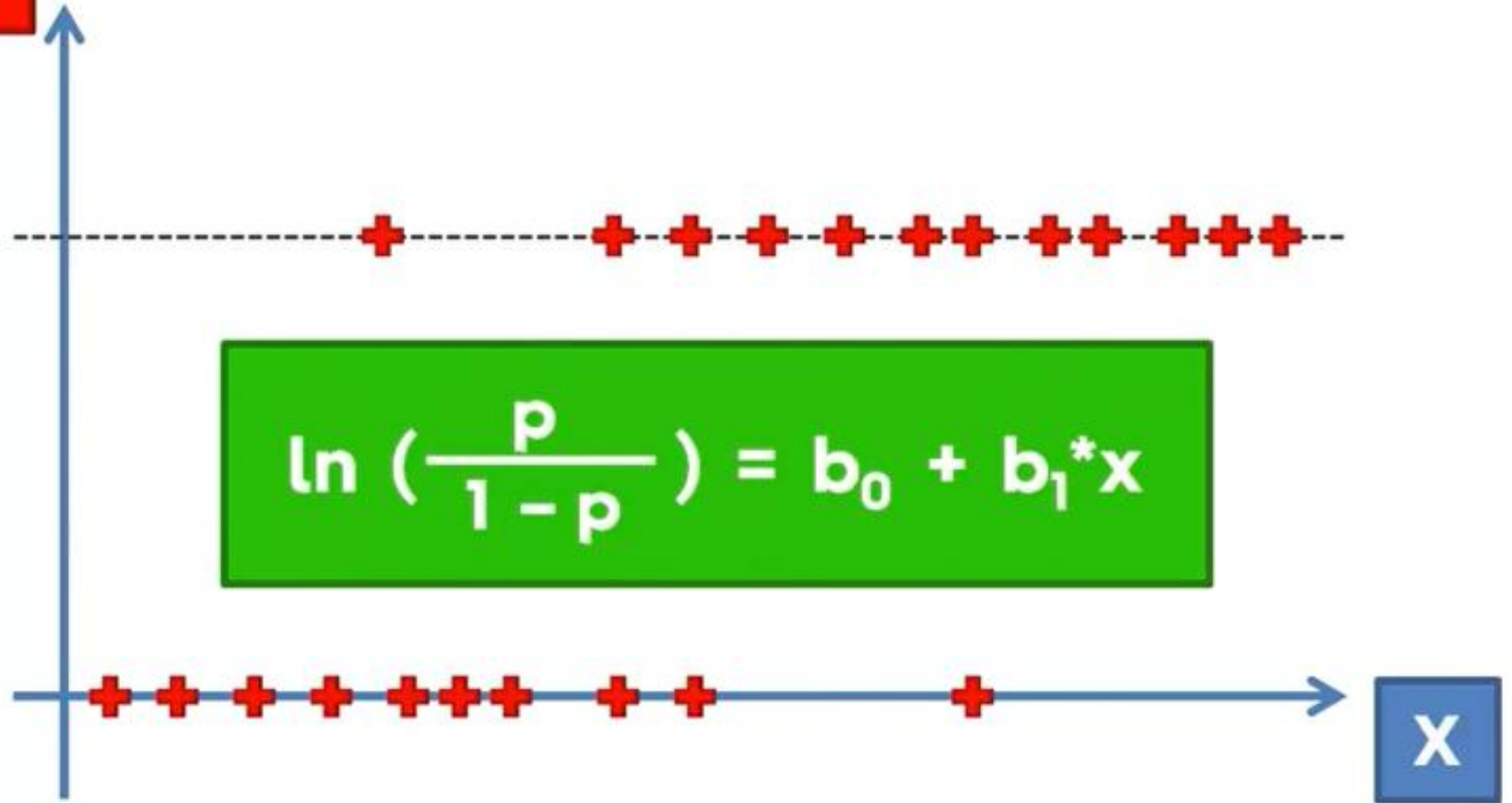
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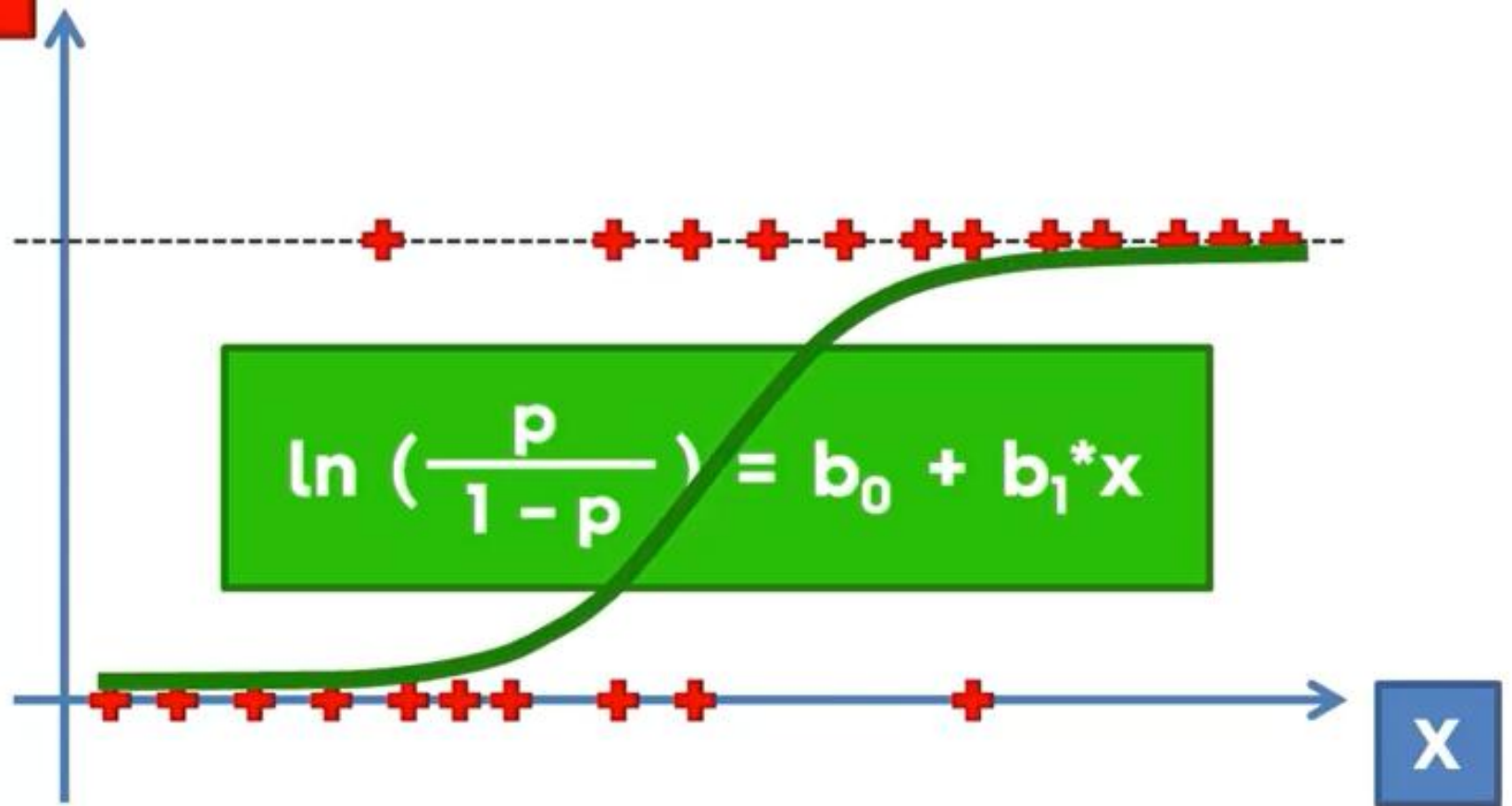
y (Actual DV)



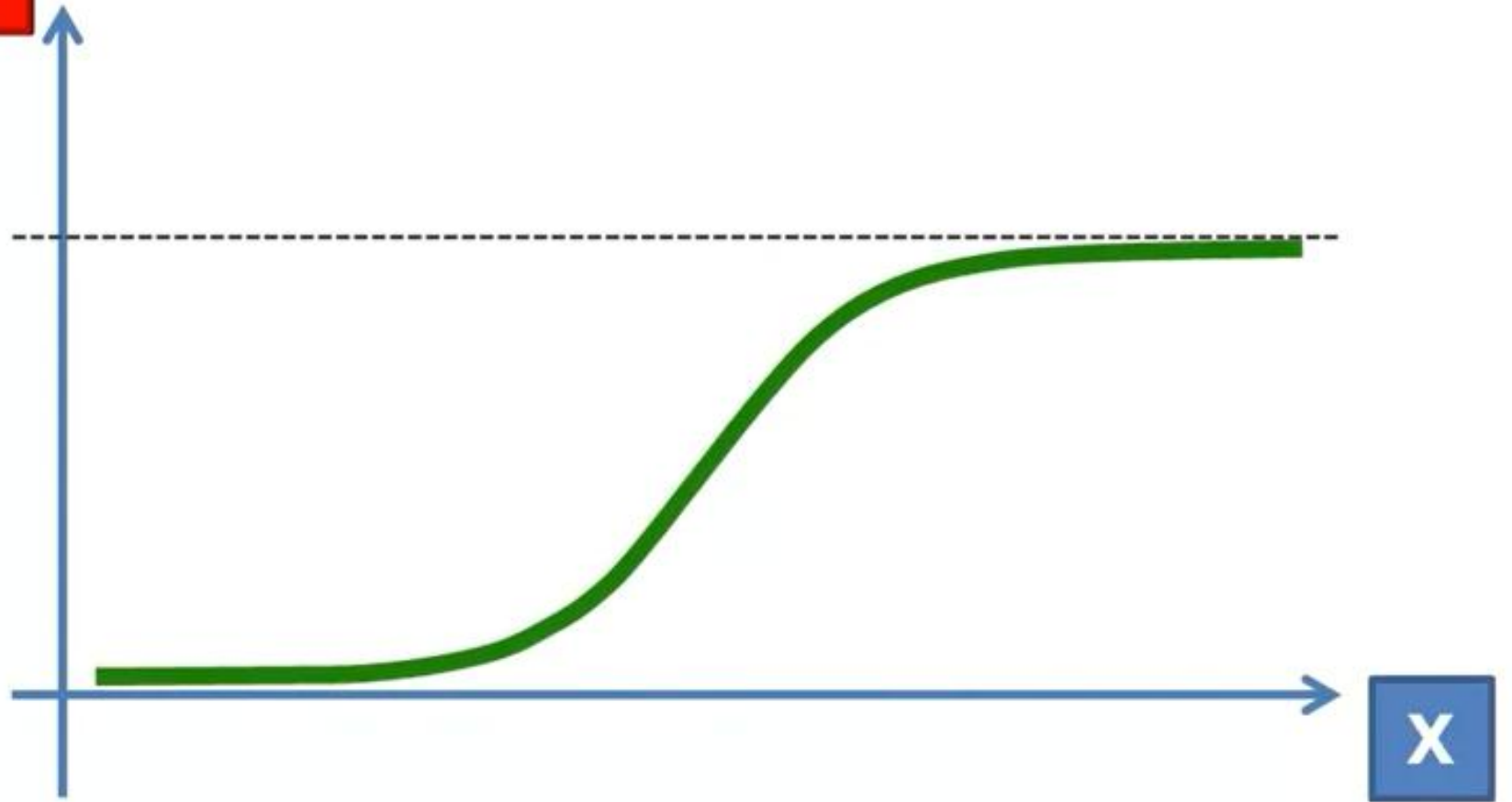
y (Actual DV)



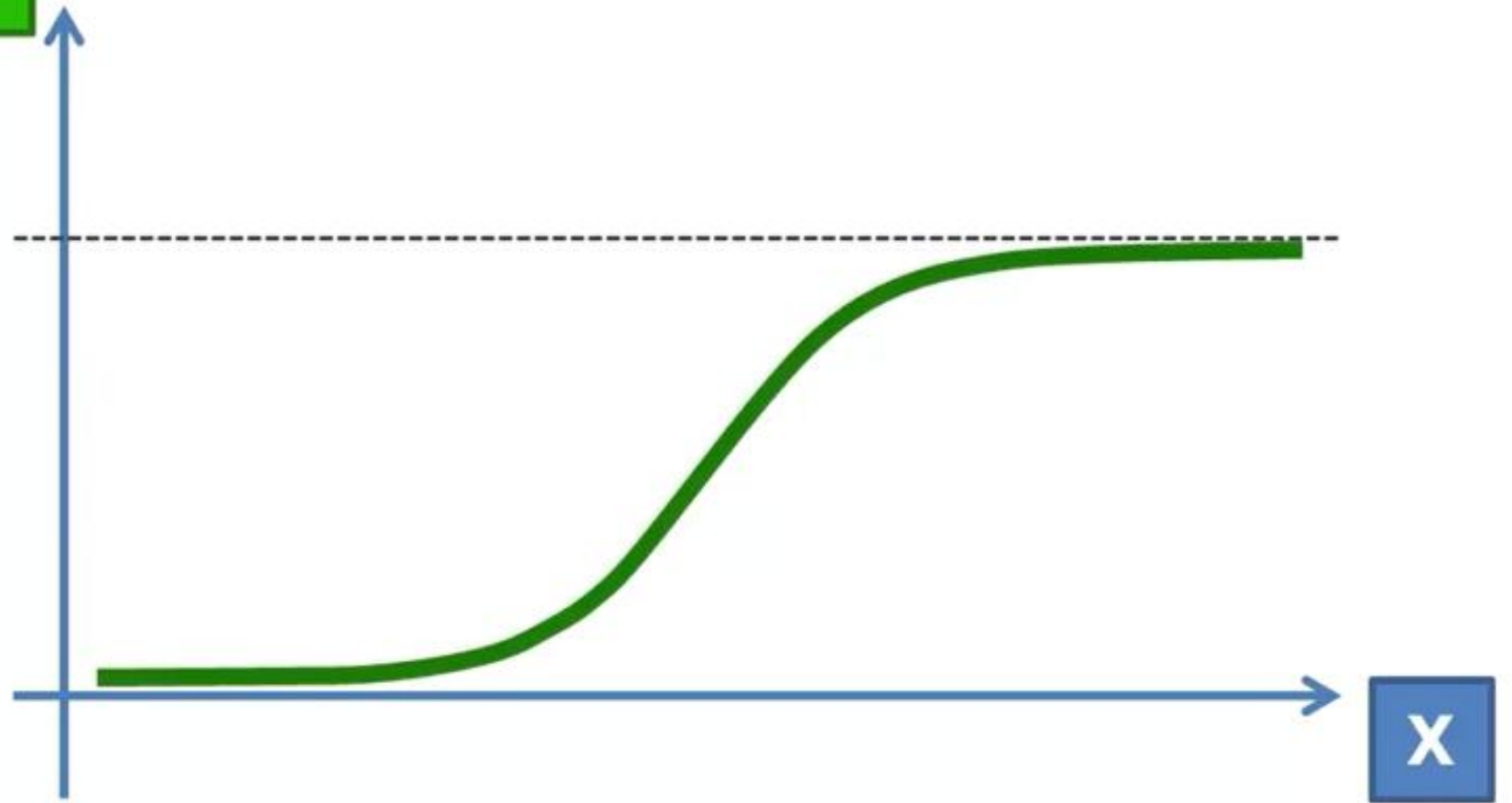
y (Actual DV)

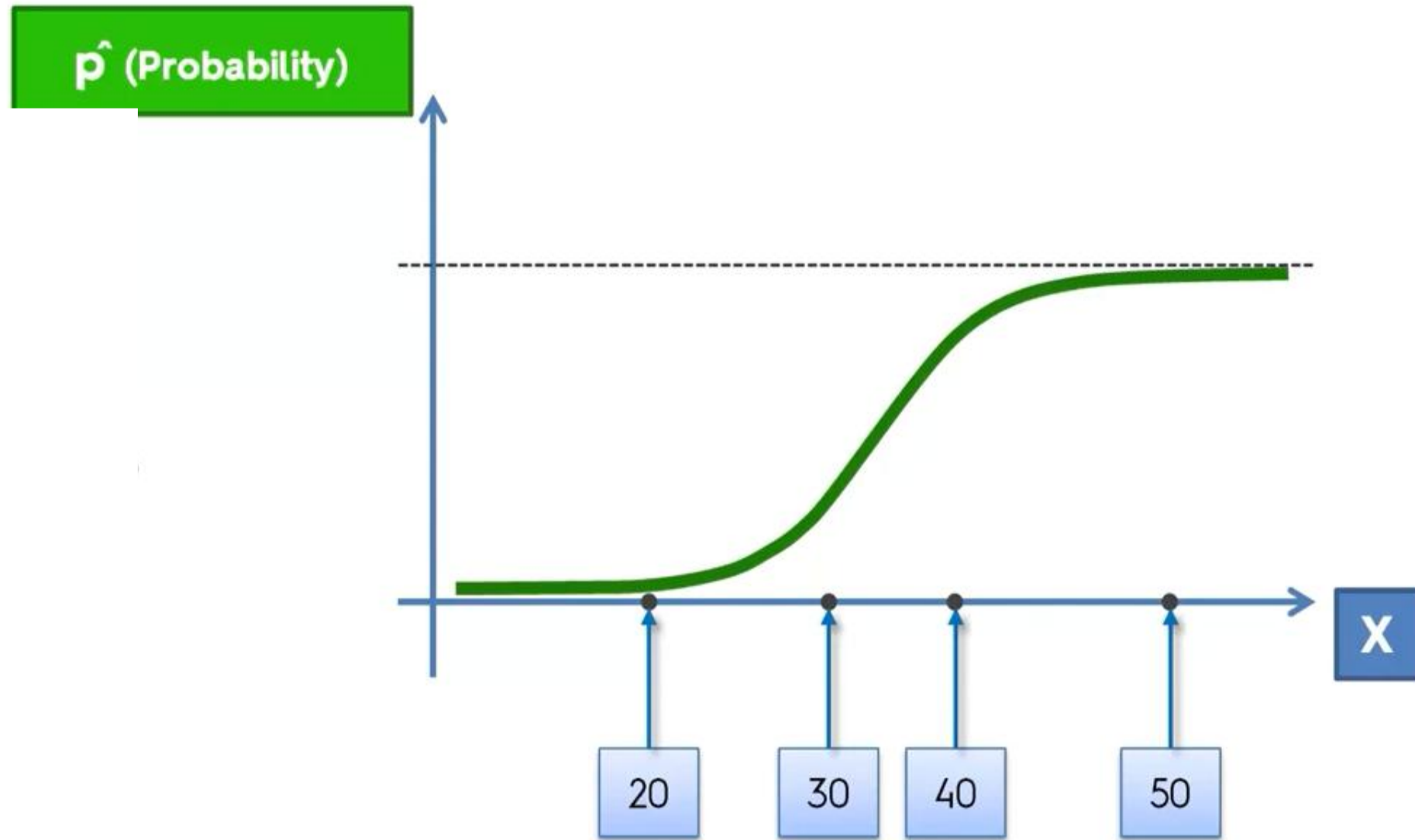


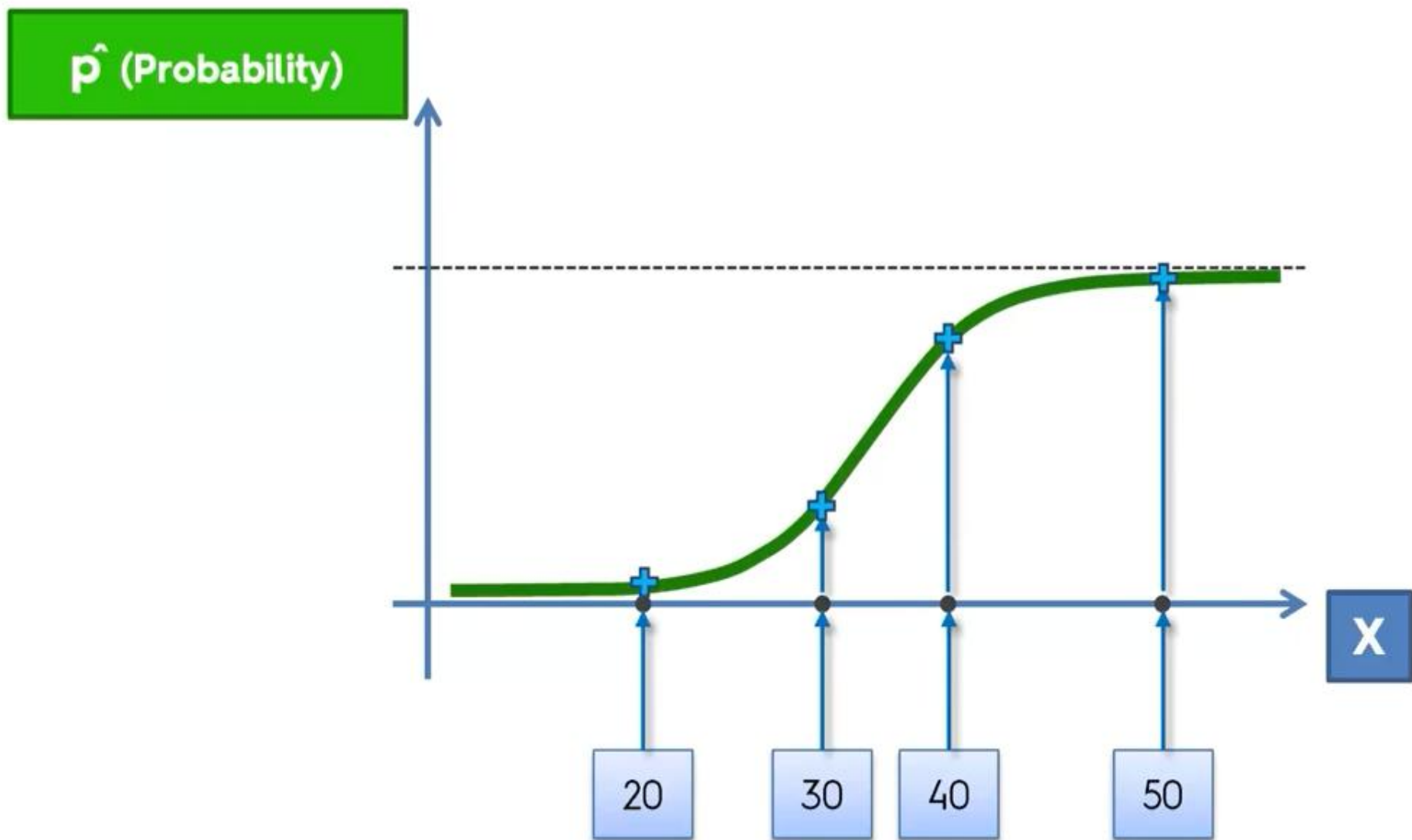
y (Actual DV)

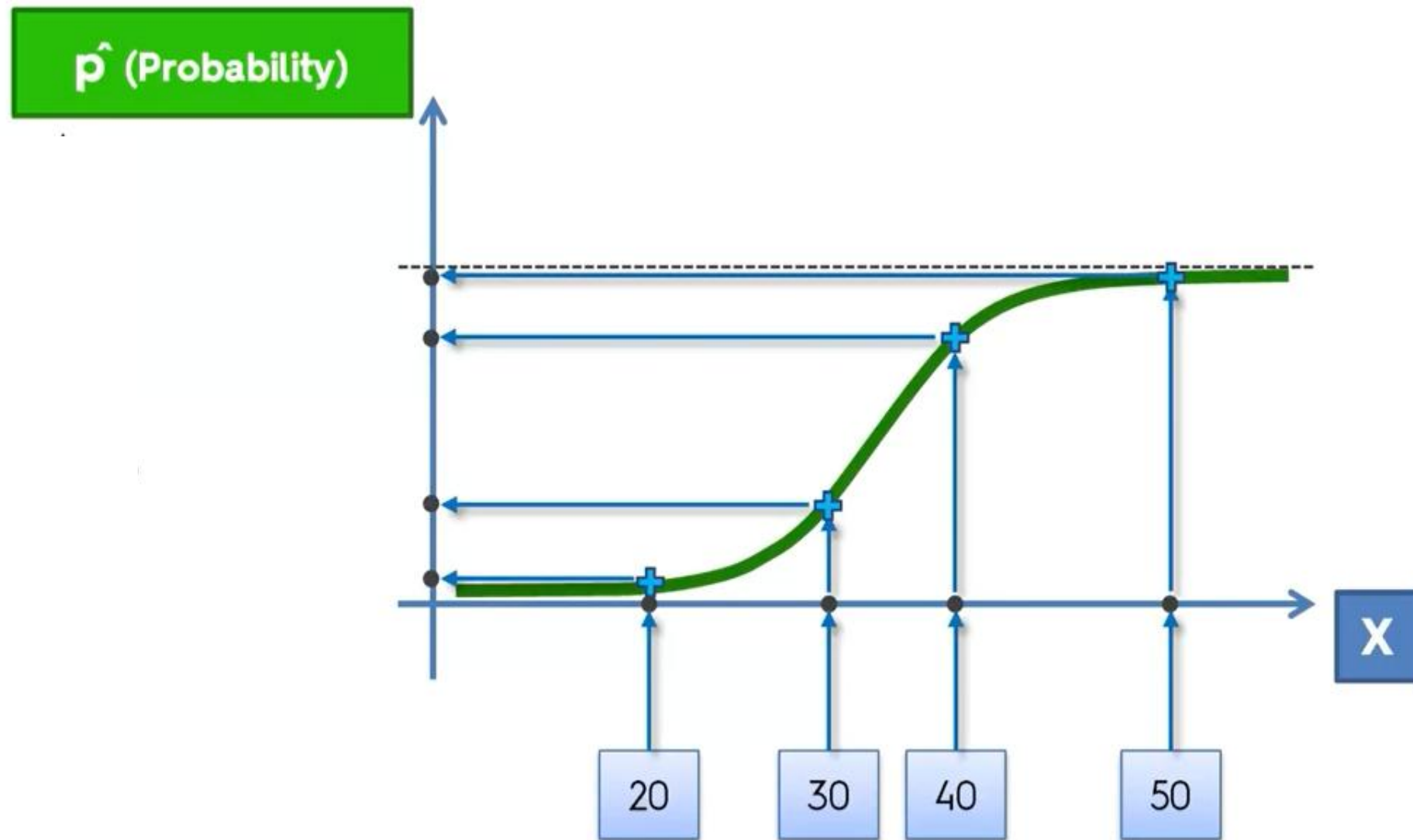


$\hat{p}$ (Probability)

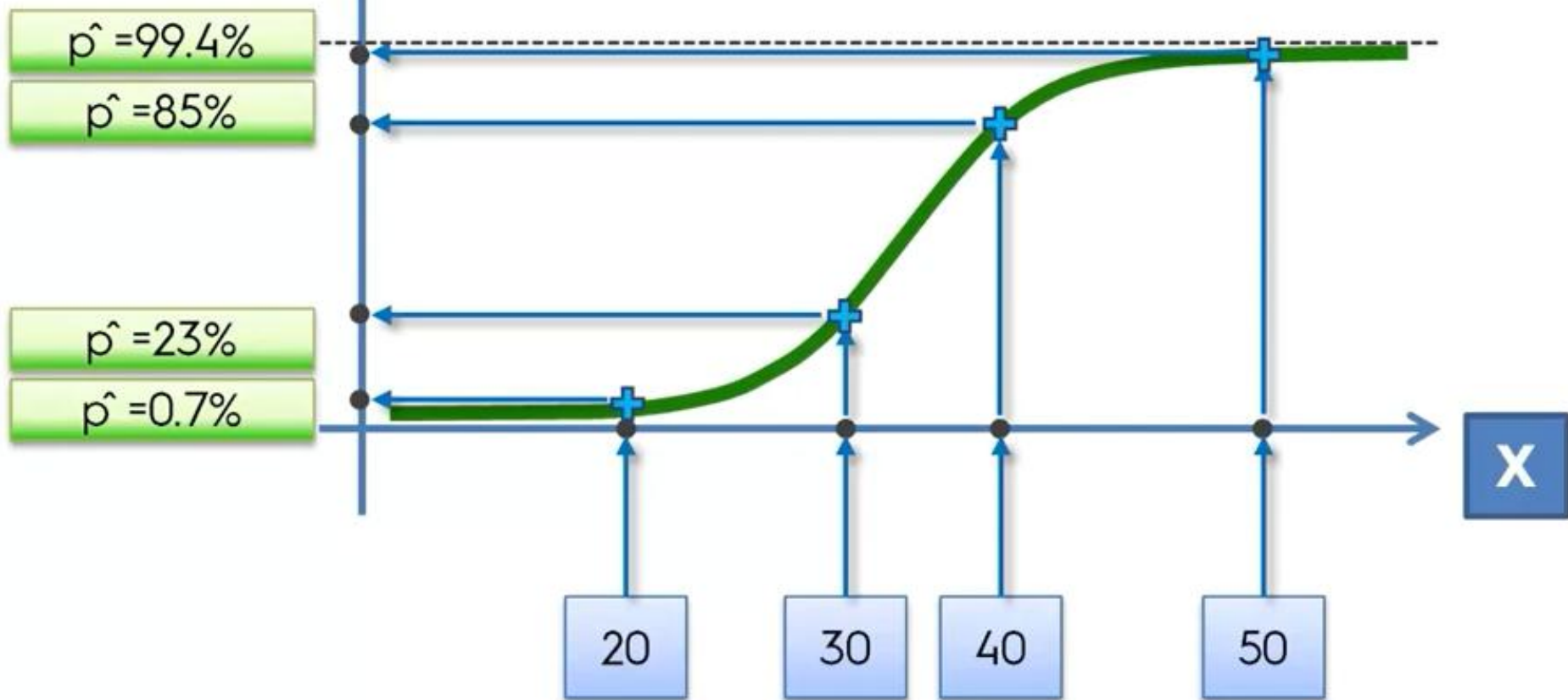




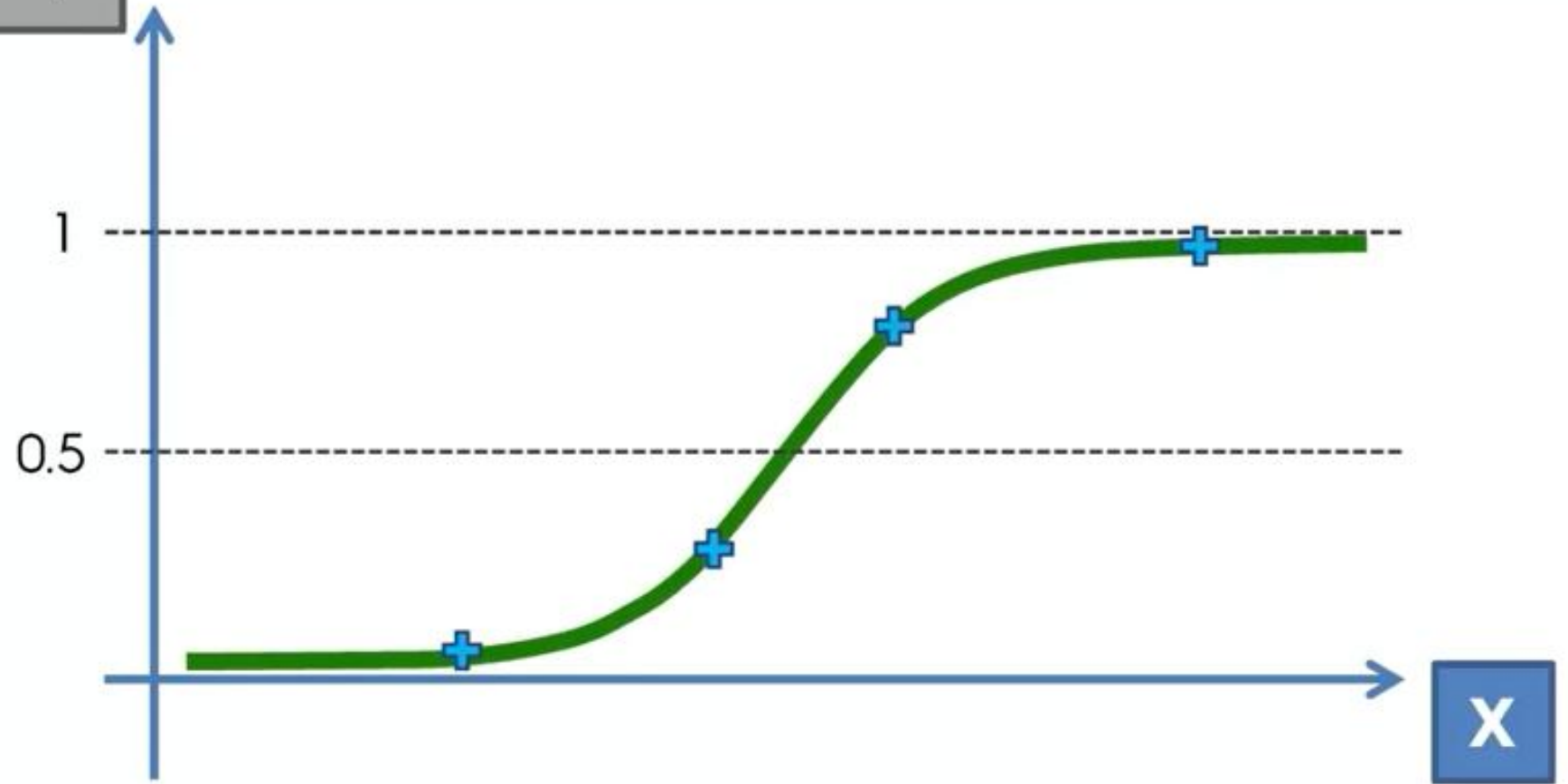




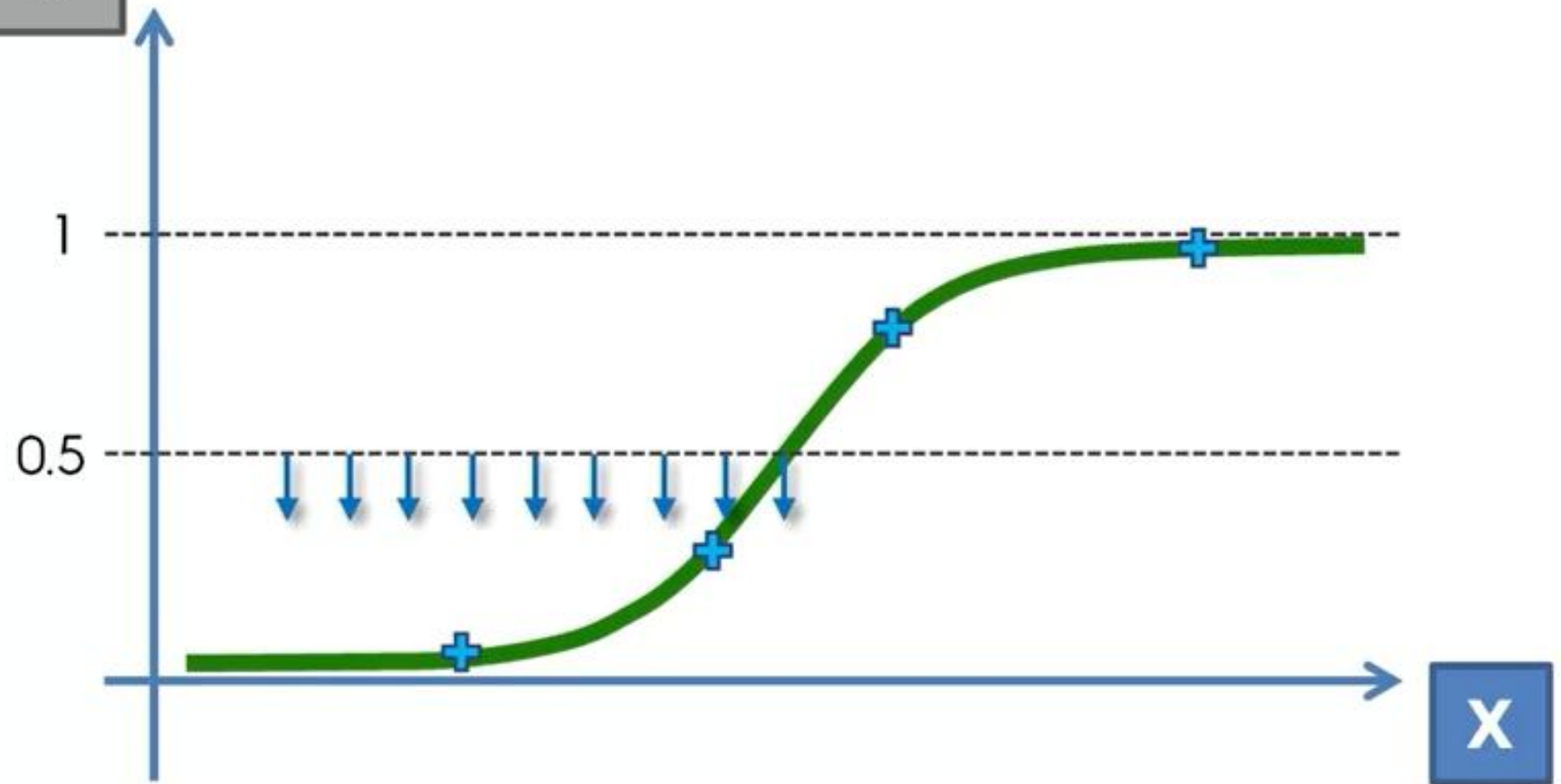
$\hat{p}$ (Probability)



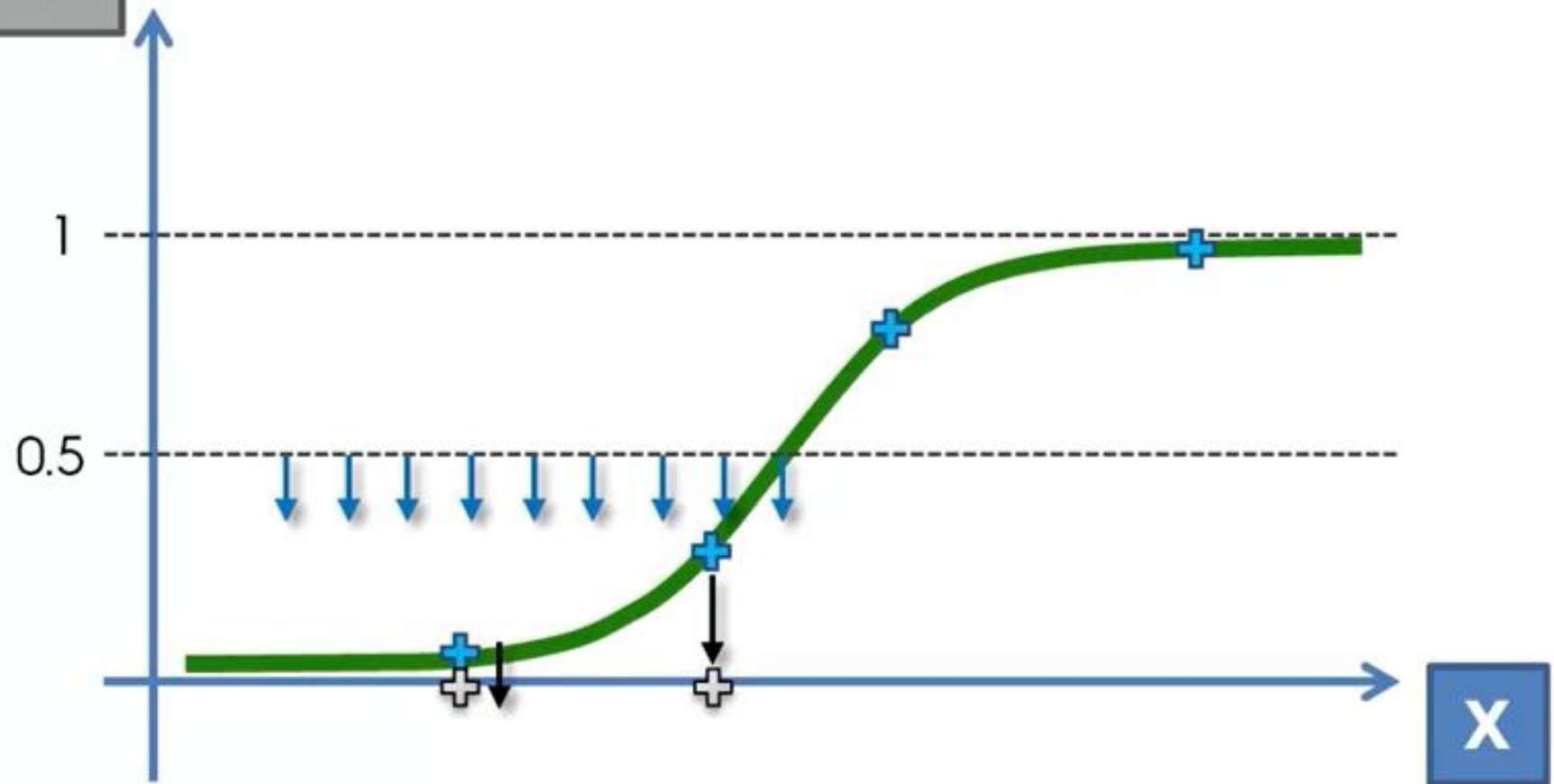
$\hat{y}$ (Predicted DV)



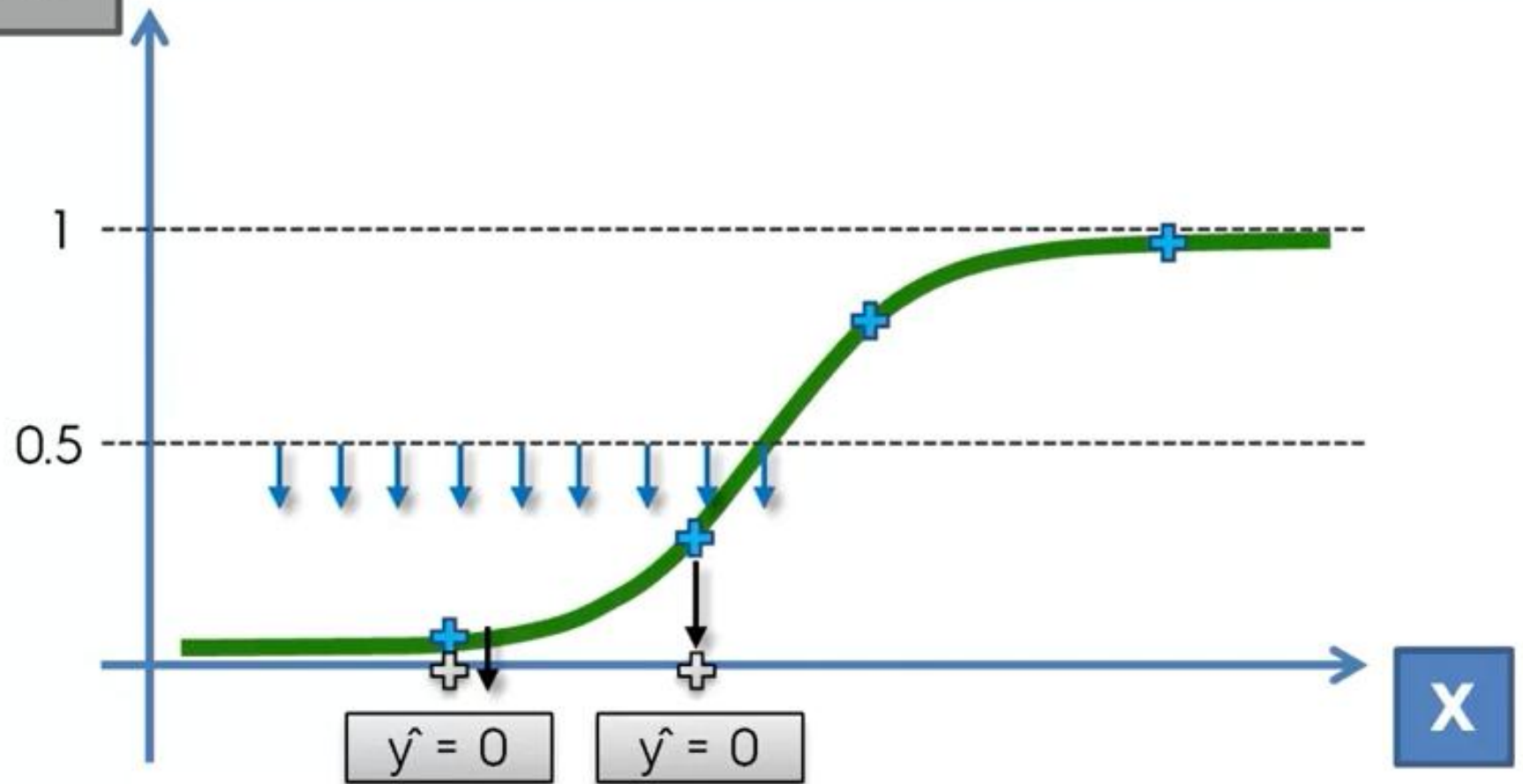
$\hat{y}$ (Predicted DV)



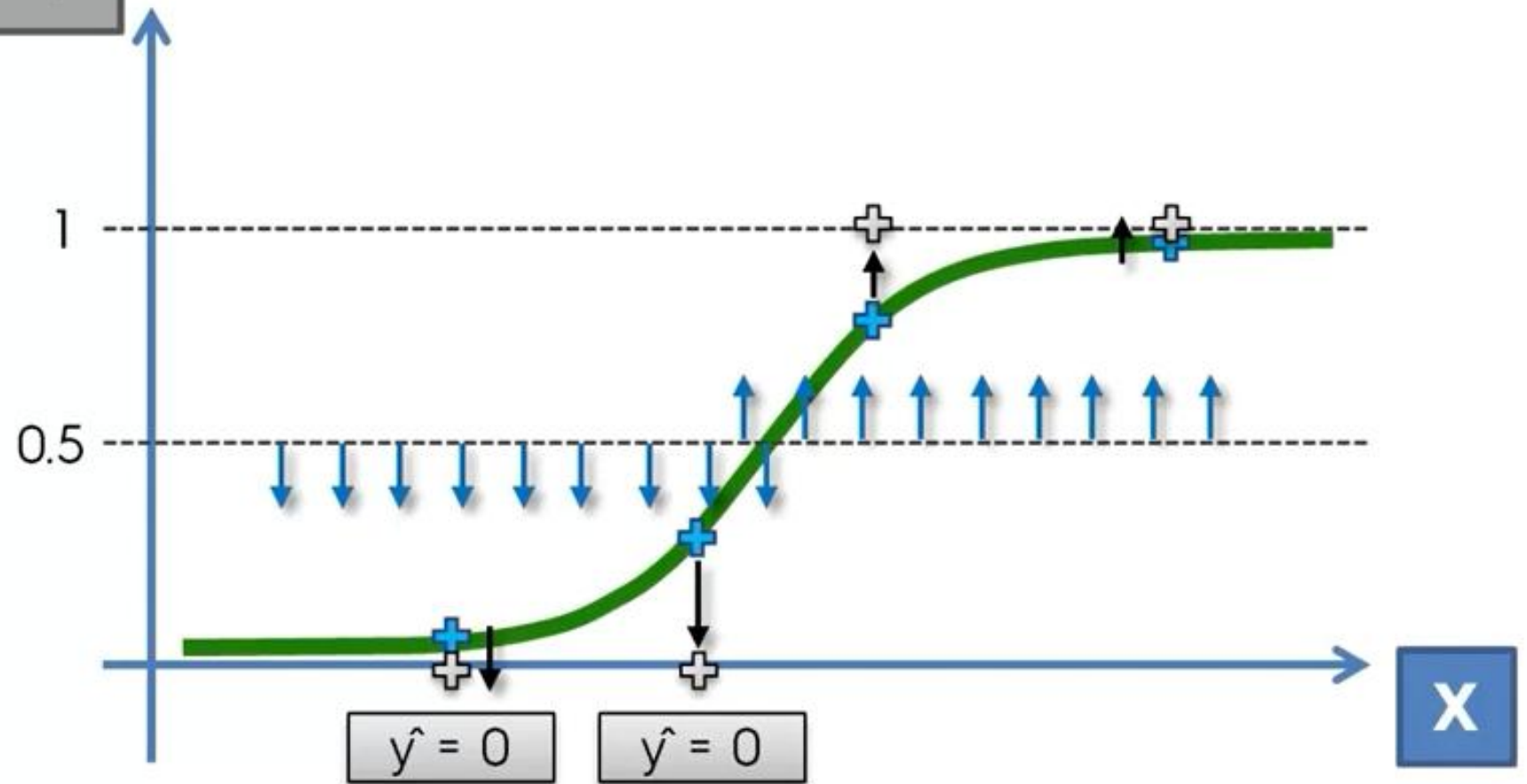
$\hat{y}$ (Predicted DV)



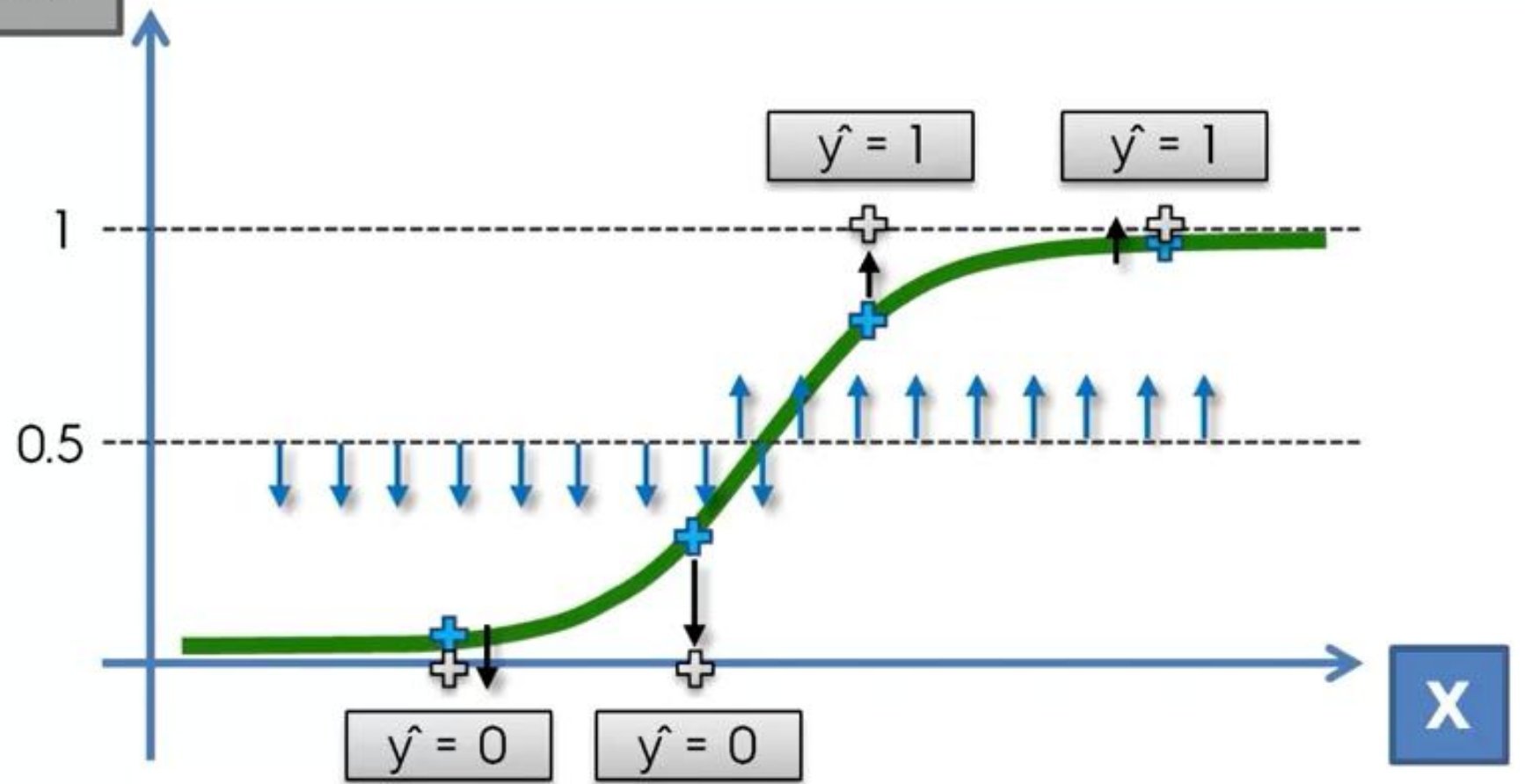
$\hat{y}$ (Predicted DV)



$\hat{y}$ (Predicted DV)



$\hat{y}$ (Predicted DV)



Logistic Regression Model

- First, the model calculates a linear score:

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$$

- Then it applies the sigmoid function:

$$P(y = 1 \mid X) = \frac{1}{1 + e^{-z}}$$

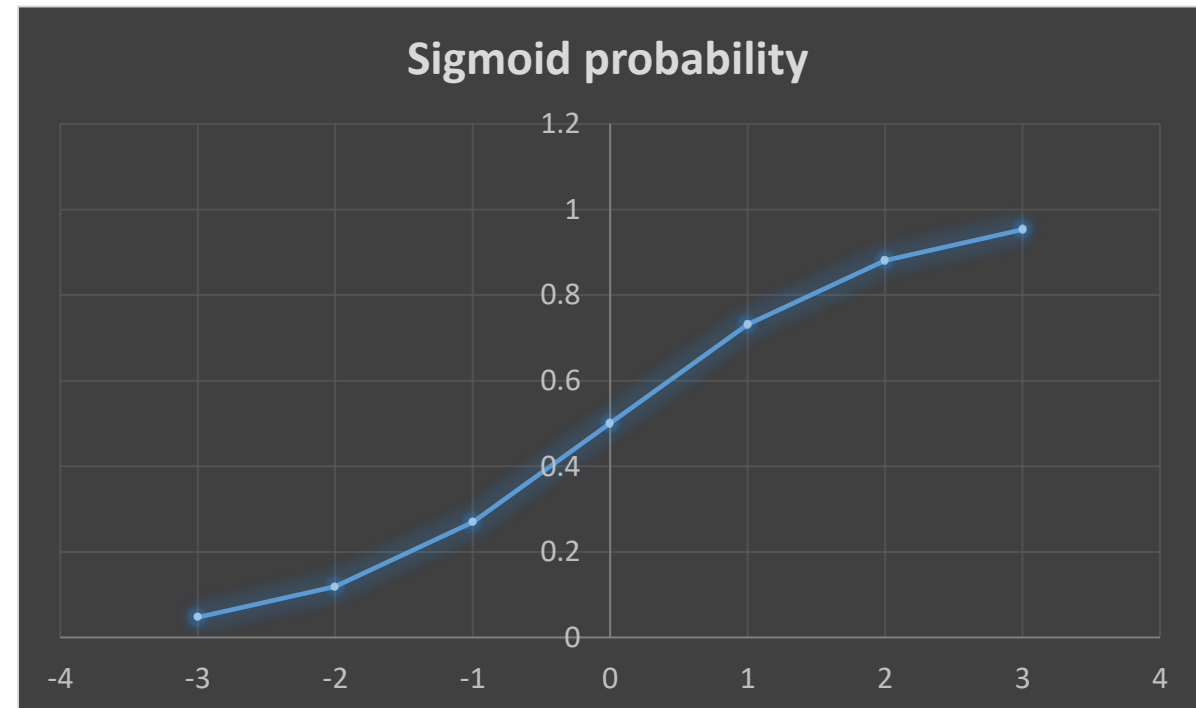
- $P(y = 1|X)$ - the probability that the output belongs to class 1, given the input features X .
- The output is always between 0 and 1.

$$P(y = 1 \mid X) = \frac{1}{1 + e^{-z}}$$

Shape of the sigmoid function

- If z is a large positive number, the probability approaches 1.
- If z is a large negative number, the probability approaches 0.
- If $z = 0$, the probability is 0.5.

z	Sigmoid probability
-3	0.047
-2	0.119
-1	0.269
0	0.500
1	0.731
2	0.881
3	0.953

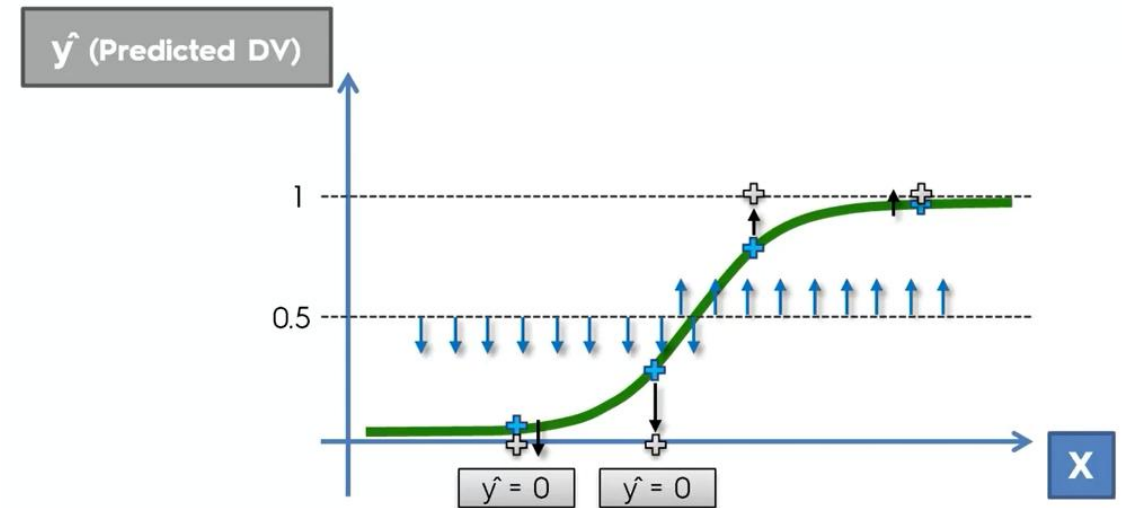


Decision Threshold

- The probability is converted into a class using a threshold.

- For a threshold of 0.5:
$$\hat{y} = \begin{cases} 1, & P(y = 1 | X) \geq 0.5 \\ 0, & P(y = 1 | X) < 0.5 \end{cases}$$

- The threshold does not always need to be 0.5.
- A **lower** threshold **increases sensitivity** or **recall**.
- A **higher** threshold **reduces false-positive** predictions.



Example 1: Single Input Feature

- Suppose a logistic regression model **predicts whether a student will pass an examination based on study hours.**
- The model is:

$$z = -4 + 0.8x$$

where x represents the number of study hours.

- Find the probability that a student who studies for 6 hours will pass.

- Step 1: Calculate the linear score

$$z = -4 + 0.8(6)$$

$$z = -4 + 4.8 = 0.8$$

- Step 2: Apply the sigmoid function

$$P(y = 1) = \frac{1}{1 + e^{-0.8}}$$

$$e^{-0.8} \approx 0.4493$$

$$P(y = 1) = \frac{1}{1 + 0.4493}$$

$$P(y = 1) \approx 0.690$$

Step 3: Predict the class

Using a threshold of 0.5:

$$0.690 \geq 0.5$$

$$\hat{y} = 1$$

The student is predicted to pass, with a probability of approximately 69%.

Example 2: Multiple Input Features

- Suppose the probability of loan approval is determined by:

$$z = -5 + 0.00005x_1 - 0.8x_2$$

where:

- x_1 = annual income
- x_2 = existing loan status
 $x_2 = 1$ if the customer has an existing loan; otherwise $x_2 = 0$
- For a customer with an annual income of ₹80,000 and an existing loan:

$$x_1 = 80000, \quad x_2 = 1$$

- Step 1: Calculate z

$$z = -5 + 0.00005(80000) - 0.8(1)$$

$$z = -5 + 4 - 0.8 = -1.8$$

- Step 2: Calculate the probability

$$P(y = 1) = \frac{1}{1 + e^{1.8}}$$

$$P(y = 1) \approx 0.142$$

- Step 3: Predict the class
Using a threshold of 0.5:

$$0.142 < 0.5$$

Therefore, the loan application is predicted to be rejected.

Odds and Log-Odds

- If the probability of an event is p , its odds are:

$$\text{Odds} = \frac{p}{1 - p}$$

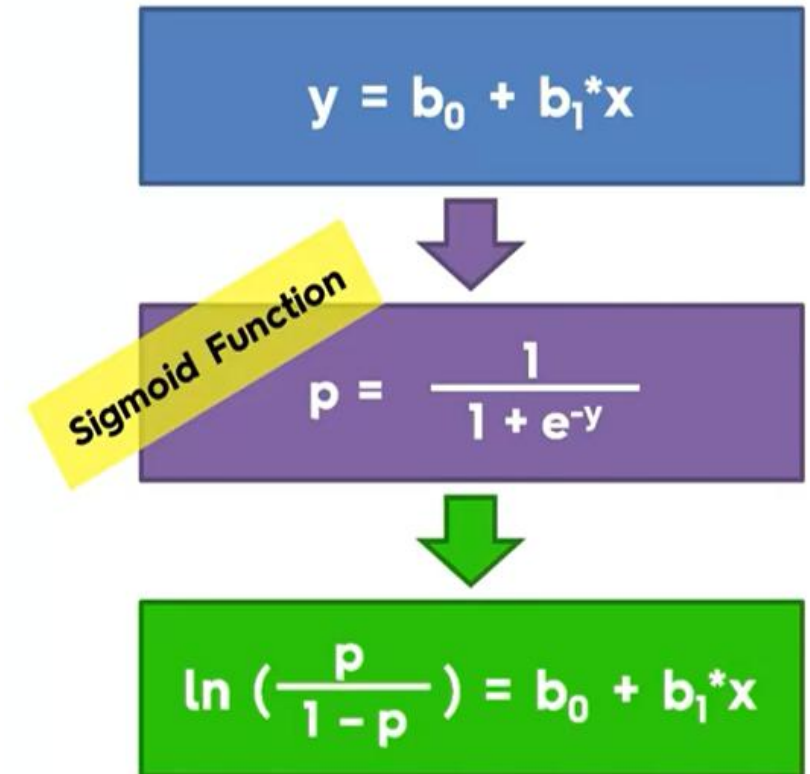
- For example, if:

$$p = 0.8$$

- then:

$$\text{Odds} = \frac{0.8}{1 - 0.8} = 4$$

- The odds are 4 to 1 in favour of the event.



- Taking the logarithm gives the log-odds or logit: $\log \left(\frac{p}{1-p} \right)$
- Logistic regression models the log-odds as a linear function:

$$\log \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

Interpreting Model Coefficients

- Consider:

$$\log \left(\frac{p}{1-p} \right) = -3 + 0.5x$$

- The coefficient of x is 0.5.
- A one-unit increase in x increases the log-odds by 0.5.
- The corresponding odds ratio is:

$$e^{0.5} \approx 1.649$$

- Therefore, for every one-unit increase in x , the odds of the positive outcome are multiplied by approximately 1.649.

- This is equivalent to an approximate 64.9% increase in the odds:

$$(1.649 - 1) \times 100 = 64.9\%$$

- This describes the change in odds, not necessarily the change in probability.

Cost Function

- **Mean squared error** is generally **not used** to train logistic regression. Instead, it uses **binary/ categorical cross-entropy**, also called **log loss**:

$$J(\beta) = -\frac{1}{n} \sum_{i=1}^n [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

where:

- y_i is the actual class.
- p_i is the predicted probability of class i .
- n is the number of observations.

- For one observation: $L = -[y \log(p) + (1 - y) \log(1 - p)]$

- Case 1: Actual class is 1 ($y = 1$)

$$L = -\log(p)$$

- If $p = 0.9$: $L = -\log(0.9) \approx 0.105$

- If $p = 0.1$: $L = -\log(0.1) \approx 2.303$

- Case 2: Actual class is 0 ($y = 0$)

$$L = -\log(1 - p)$$

$$\text{If } p = 0.9 \quad L \approx 2.303$$

$$\text{If } p = 0.1 \quad L \approx 0.105$$

A confident incorrect prediction receives a large penalty.

Example 3: Calculate Log Loss

- Consider the following results:

Observation	Actual class y	Predicted probability p
1	1	0.9
2	0	0.2
3	1	0.7

- The average log loss is:

$$J = -\frac{1}{3} \sum_{i=1}^3 [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

$$J = -\frac{1}{3}[\log(0.9) + \log(0.8) + \log(0.7)]$$

$$J = -\frac{1}{3}[-0.1054 - 0.2231 - 0.3567]$$

$$J = \frac{0.6852}{3}$$

$$J \approx 0.2284$$

A **smaller log-loss** value indicates **better probability predictions**.

How Logistic Regression Learns the Coefficients

- The model learns $\beta_0, \beta_1, \dots, \beta_p$ by minimizing **log loss**.
- A common optimization method is **gradient descent**:

$$\beta_j := \beta_j - \alpha \frac{\partial J}{\partial \beta_j}$$

- For logistic regression:

$$\frac{\partial J}{\partial \beta_j} = \frac{1}{n} \sum_{i=1}^n (p_i - y_i) x_{ij}$$

where α is the learning rate.

$$\beta_j := \beta_j - \alpha \left[\frac{1}{n} \sum_{i=1}^n (p_i - y_i) x_{ij} \right]$$

Decision Boundary

- For a threshold of 0.5: $P(y = 1) = 0.5$
- The sigmoid probability is 0.5 when: $\mathbf{z}=\mathbf{0}$

z	Sigmoid probability
-3	0.047
-2	0.119
-1	0.269
0	0.500
1	0.731
2	0.881
3	0.953

- For a model with two features: $z = \beta_0 + \beta_1 x_1 + \beta_2 x_2$
- The decision boundary is:

$$\beta_0 + \beta_1 x_1 + \beta_2 x_2 = 0$$

$$x_2 = -\frac{\beta_0 + \beta_1 x_1}{\beta_2}$$

Basic logistic regression produces a **linear decision** boundary.

Nonlinear boundaries can be obtained by adding **polynomial or transformed features**.

Changing the Classification Threshold

Increasing the threshold generally:

- Reduces positive predictions
- Reduces false positives
- Increases precision
- May decrease recall

Decreasing the threshold generally:

- Increases positive predictions
- Reduces false negatives
- Increases recall
- May decrease precision

		ACTUAL VALUES	
		POSITIVE	NEGATIVE
PREDICTED VALUES	POSITIVE	560	60
	NEGATIVE	50	330

550	50
60	340

Increasing the threshold generally:

- Reduces positive predictions
- Reduces false positives
- Increases precision
- May decrease recall

$$Precision = \frac{TP}{TP + FP} = \frac{560}{560 + 60} = 0.903$$

$$Sensitivity \text{ or } Recall = \frac{TP}{TP + FN} = \frac{560}{560 + 50} = 0.918$$

$$Precision = \frac{550}{550 + 50} = 0.9166 \text{ (inc)}$$

$$Recall = \frac{550}{550 + 60} = 0.9016 \text{ (dec)}$$

Decreasing the threshold generally:

- Increases positive predictions
- Reduces false negatives
- Increases recall
- May decrease precision

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{560}{560 + 60} = 0.903$$

$$\text{Sensitivity or Recall} = \frac{TP}{TP + FN} = \frac{560}{560 + 50} = 0.918$$

570	70
30	330

$$\text{Precision} = \frac{570}{570 + 70} = 0.8906 \text{ (dec)}$$

$$\text{Recall} = \frac{570}{550 + 30} = 0.95 \text{ (inc)}$$

Regularization

- Regularization helps prevent overfitting by penalizing large coefficients.

L2 regularization

$$J_{L2} = J + \lambda \sum_{j=1}^p \beta_j^2$$

- L2 regularization reduces coefficients but usually does not make them exactly zero.

- L1 regularization

$$J_{L1} = J + \lambda \sum_{j=1}^p |\beta_j|$$

- L1 regularization can make some coefficients exactly zero. It can therefore perform feature selection.

Cost function

In scikit-learn:

$$C = \frac{1}{\text{regularization strength}}$$

Therefore:

- Small C : stronger regularization
- Large C : weaker regularization

Multiclass Logistic Regression

- When there are more than two classes, logistic regression can estimate the probability of each class.
- For class (k):

$$P(y = k \mid X) = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}}$$

- This is known as the **softmax** function.

Advantages

- Simple to implement
- Computationally efficient
- Produces class probabilities
- Coefficients are interpretable
- Works well for approximately linear decision boundaries
- Can be regularized
- Provides a strong baseline classifier

Limitations

- Basic logistic regression produces a linear decision boundary.
- It may perform poorly when relationships are highly nonlinear.
- It can be affected by multicollinearity.
- Outliers can influence coefficients.
- Performance may be misleading on imbalanced datasets if only accuracy is used.

Logistic Regression vs Linear Regression

Characteristic	Linear regression	Logistic regression
Main purpose	Predict continuous values	Predict classes
Output	Any real number	Probability between 0 and 1
Function	Linear function	Sigmoid applied to a linear score
Loss function	Mean squared error	Log loss
Example	Predict salary	Predict pass/fail

Solve

Given the model: $\mathbf{z} = -6 + 1.2 \mathbf{x}$

Where (x) is the number of study hours:

- Calculate the probability of passing for (x=3).
- Calculate the probability of passing for (x=5).
- Calculate the probability of passing for (x=7).

Predict the class using a threshold of 0.5.

Repeat using a threshold of 0.7.

Thank you